GLOBAL EDITION



MICROECONOMICS

14th Edition

Michael Parkin



1 WHAT IS ECONOMICS?

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After studying this chapter, you will be able to:

- Define economics and distinguish between microeconomics and macroeconomics
- Explain the two big questions of economics
- Explain the key ideas that define the economic way of thinking
- Explain how economists go about their work as social scientists and policy advisers
- Describe the jobs available to a graduate with a major in economics

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Definition of Economics

All economic questions arise because we want more than we can get.

Our inability to satisfy all our wants is called **scarcity**.

Because we face scarcity, we must make ***choices***.

The choices we make depend on the incentives we face.

An **incentive** is a reward that encourages an action or a penalty that discourages an action.

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Definition of Economics

Economics is the social science that studies the *choices* that individuals, businesses, governments, and entire societies make as they cope with *scarcity* and the *incentives* that influence and reconcile those choices.

Economics divides in two main parts:

- Microeconomics
- Macroeconomics

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→ the concept of economics defines (scarce in economic terms) limited resources and unlimited wants which requires us to make choices

→ the most effective choice should return maximum satisfaction to us

→ optimal satisfaction requires benefits to exceed costs

e.g. the benefit of learning by attending a lecture at the cost of consuming time that could've been spent on other activities.

Differentiating microeconomics and macroeconomics

Macroeconomics is a broader economy studied with respect to specific variables e.g. GDP, inflation rate, etc.

Microeconomics is more niche and focuses on individual behavior e.g. analyzing one local bank in Riyadh

Definition of Economics

↑ seller/buyer behavior as a single body

Microeconomics is the study of choices that individuals and businesses make, the way those choices interact in markets, and the influence of governments.

Two microeconomic questions:
Why are people streaming more movies?
Would a tax on online shopping effect Amazon?

Macroeconomics is the study of the performance of the national and global economies.

Two macroeconomic questions:
Why does the unemployment rate fluctuate?
Can the Federal Reserve make the unemployment rate fall by keeping interest rates low?

examples

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Two Big Economic Questions

Two big questions summarize the scope of economics:

resources - understanding what to produce based on the public need

- How do choices end up determining ***what***, ***how***, and ***for whom*** goods and services get produced?
- When do choices made in the pursuit of ***self-interest*** also promote the ***social interest***?

decision working in one's individual favor

considering society
e.g. helping the economy
if we don't buy anything
the economy will crash

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Two Big Economic Questions

What, How, and For Whom?

Goods and services are the objects that people value and produce to satisfy human wants.

What?

In the United States, agriculture accounts for less than 1 percent of total production, industry (manufactured goods) for 11 percent, and services for 80 percent.

In low-income Ethiopia, agriculture accounts for 35 percent of total production, industry goods for 22 percent, and services for 44 percent.

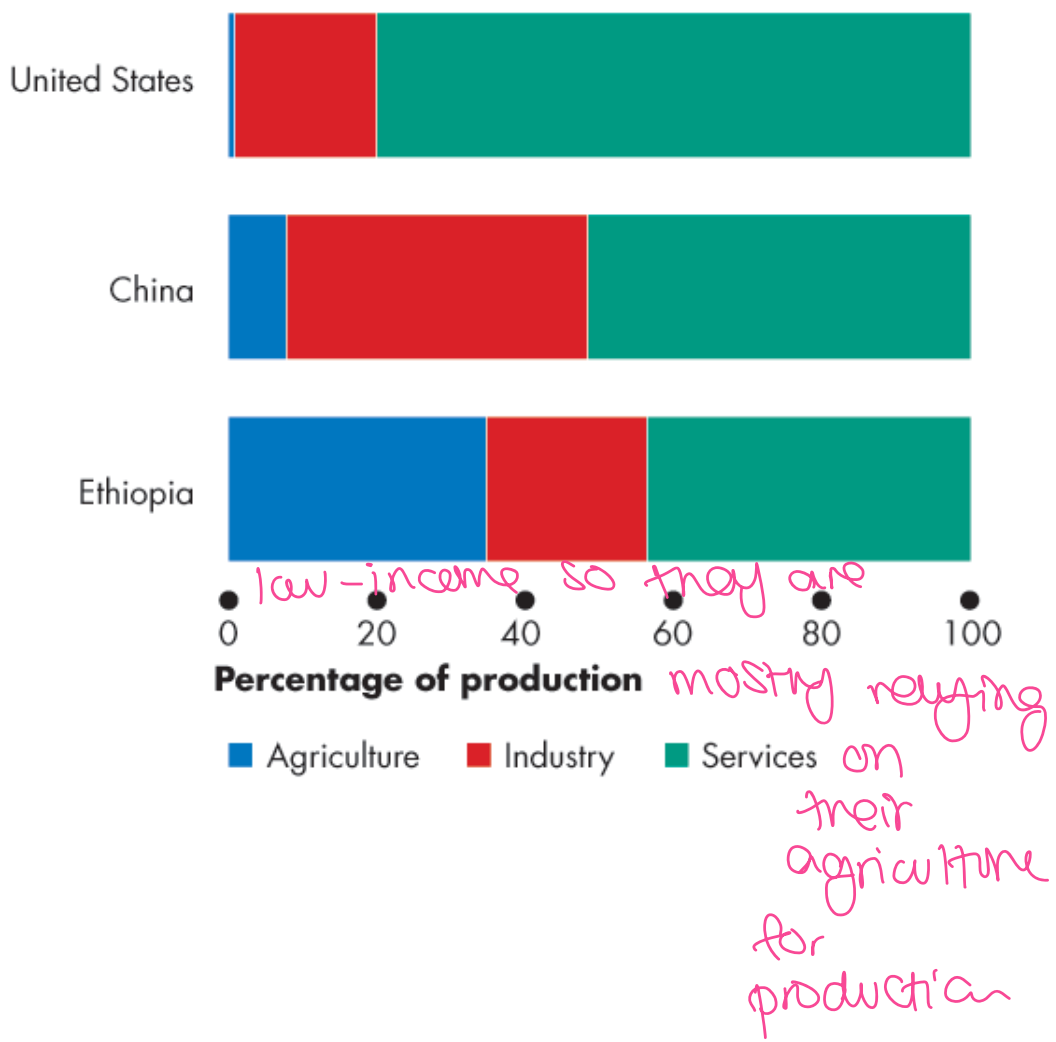
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Two Big Economic Questions

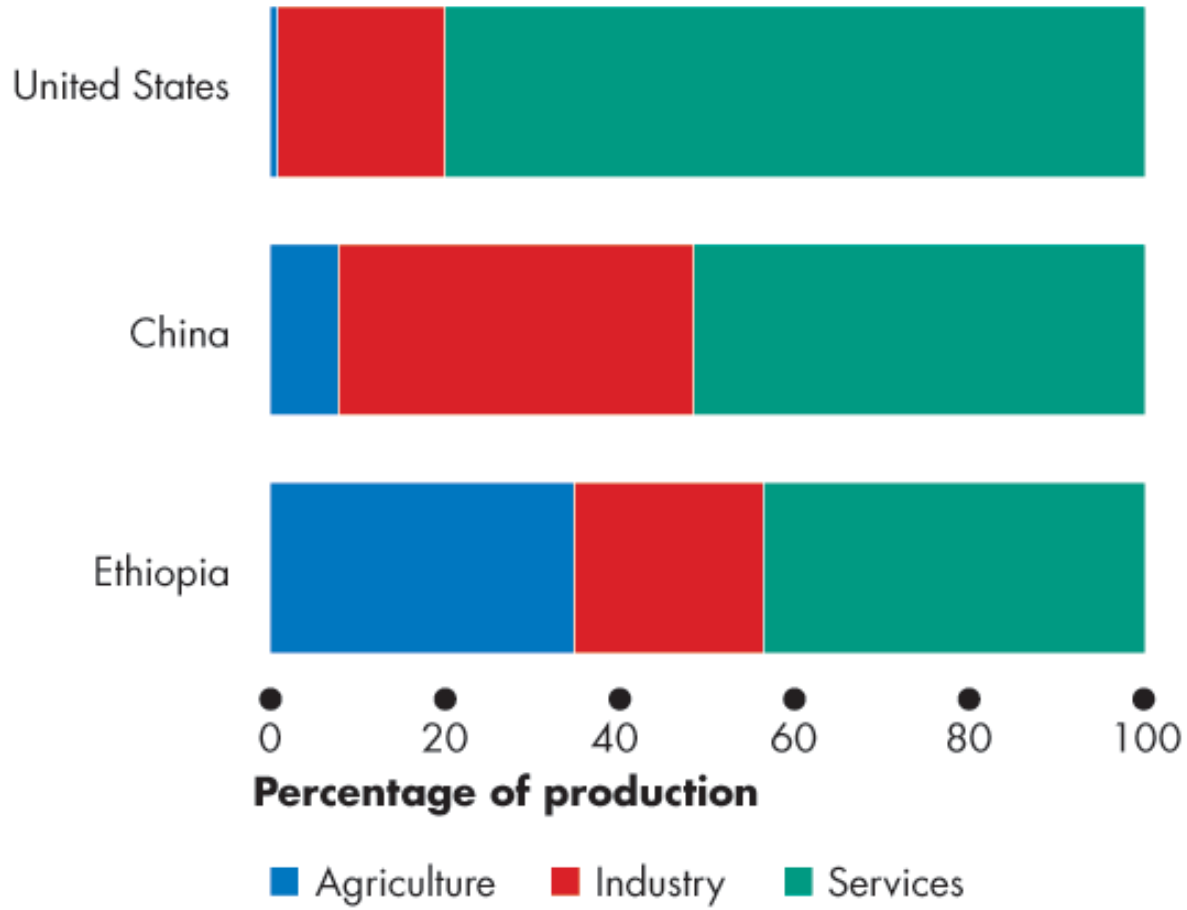
Figure 1.1 shows these numbers for the United States, China, and Ethiopia.

What determines these patterns of production?

How do choices end up determining the quantity of each item produced in the United States and around the world?



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Two Big Economic Questions

How?

Goods and services are produced by using productive resources that economists call **factors of production**.

Factors of production are grouped into four categories:

1. Land
2. Labor
3. Capital
4. Entrepreneurship

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Two Big Economic Questions

★ MUST know in detail

The "gifts of nature" that we use to produce goods and services are **land**.

The work time and work effort that people devote to producing goods and services is **labor**.

The *quality* of labor depends on **human capital**, which is the knowledge and skill that people obtain from education, on-the-job training, and work experience.

↑ working skills

The tools, instruments, machines, buildings, and other constructions that businesses use to produce goods and services are **capital**. (aka physical capital)

The human resource that organizes land, labor, and capital is **entrepreneurship**.

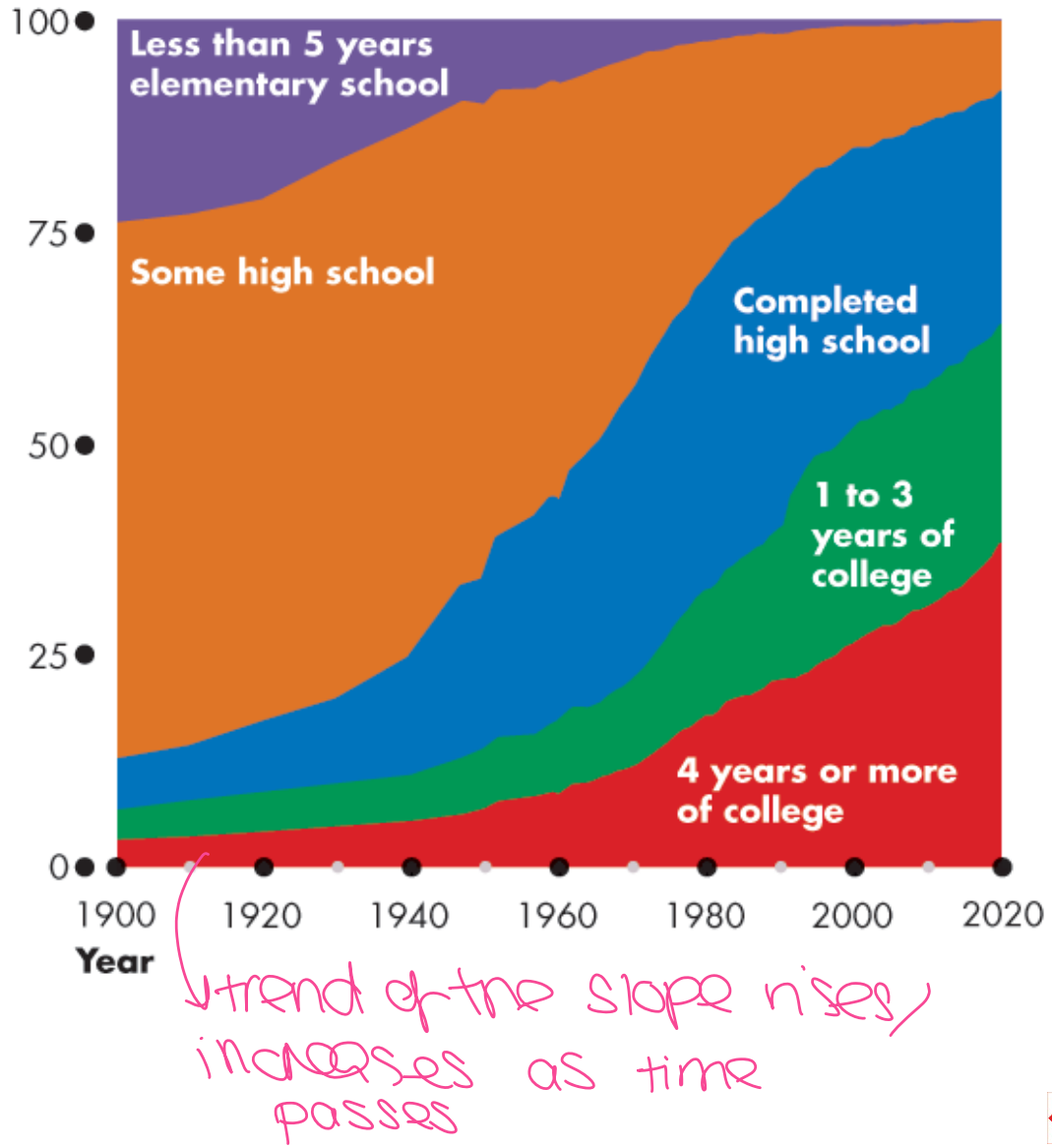
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Two Big Economic Questions

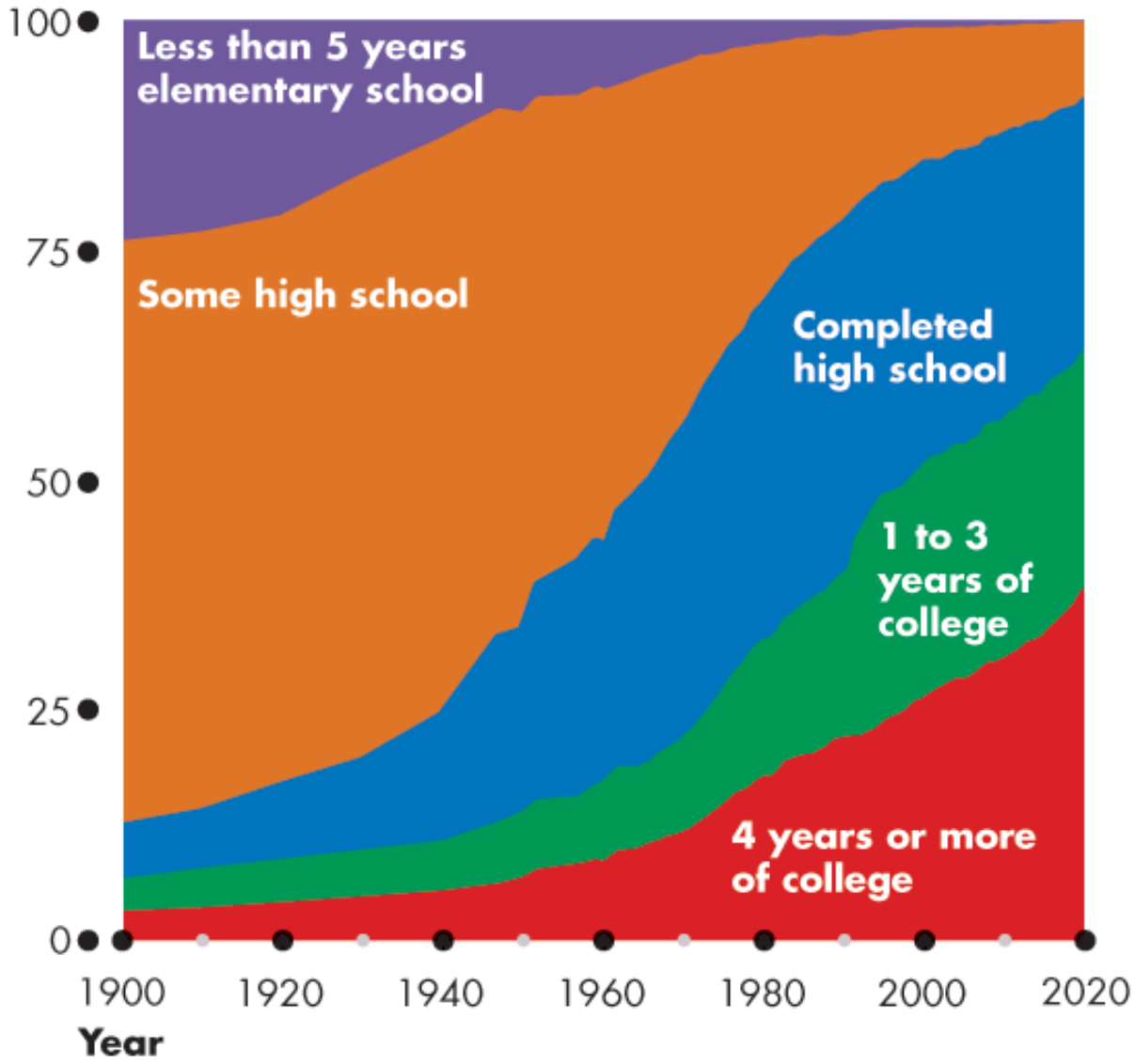
Figure 1.2 shows a measure of the growth of human capital in the United States since 1900:

The percentage of the population that has completed different levels of education.

Economics explains these trends.



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Two Big Economic Questions

For Whom?

Who gets the goods and services depends on the incomes that people earn.

- Land earns **rent**.
- Labor earns **wages**.
- Capital earns **interest**.
- Entrepreneurship earns **profit**.

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Two Big Economic Questions

Do Choices Made in the Pursuit of Self-Interest also Promote the Social Interest?

Every day, 328 million Americans and 7.9 billion people in other countries make economic choices that result in *what*, *how*, and *for whom* goods and services are produced.

These choices are made by people who are pursuing their self-interest.

Are they promoting the social interest?

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Two Big Economic Questions

Self-Interest

You make choices that are in your **self-interest**—choices that you think are best for you.

Social Interest

Choices that are best for society as a whole are said to be in the **social interest**.

Social interest has two dimensions: efficiency and fair shares.

↳ using the right resources to boost productivity

↳ isn't obtained despite efficiency due to division of the income classes

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Two Big Economic Questions

Efficiency and Social Interest

Resource use is **efficient** if it is *not* possible to make someone better off without making someone else worse off.

Fair Shares and Social Interest

The idea that the social interest requires “fair shares” is a deeply held one.

But what is a *fair share*?

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Two Big Economic Questions

Questions about the social interest are hard ones to answer and they generate discussion, debate, and disagreement.

Four topics that generate discussion and that illustrate tension between self-interest and social interest are:

- Globalization
- Information-age monopolies
- Climate change
- The Covid pandemic

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open-econ-open for trade & vice versa

Two Big Economic Questions

Globalization

trade is a crucial pillar

Globalization means the expansion of international trade, borrowing and lending, and investment.

Globalization is in the self-interest of consumers who buy low-cost imported goods and services.

↳ products & services purchased in foreign countries that are relatively cheaper than locally produced alternatives

Globalization is also in the self-interest of the multinational firms that produce in low-cost regions and sell in high-price regions. all big comp do their major production in low-income countries

But is globalization in the self-interest of low-wage workers in other countries and U.S. firms that can't compete with low-cost imports? No

Is globalization in the social interest? Yes

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Two Big Economic Questions

Information-Age Monopolies

The technological change of the past forty years has been called the *Information Revolution*.

The information revolution has clearly served your self-interest: It has provided your smartphone, laptop, loads of handy applications, and the Internet.

It has also served the self-interest of Bill Gates of Microsoft and Gordon Moore of Intel, both of whom have seen their wealth soar.

But did the information revolution serve the social interest?

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Two Big Economic Questions

Climate Change

Climate change is a huge political issue today.

Every serious political leader is acutely aware of the problem and of the popularity of having proposals that might lower carbon emissions.

Burning fossil fuels to generate electricity and to power airplanes, automobiles, and trucks pours a staggering 28 billion tons—4 tons per person—of carbon dioxide into the atmosphere each year.

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Two Big Economic Questions

Two thirds of the world's carbon emissions comes from the United States, China, the European Union, Russia, and India.

The fastest growing emissions are coming from India and China.

The amount of global warming caused by economic activity and its effects are uncertain, but the emissions continue to grow and pose huge risks.

increased production comes at the expense of polluting public's health

e.g. travelling in different cars inc. carbon emission in which it can be minimized through pooling

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Two Big Economic Questions

Every day, when you make self-interested choices to use electricity and gasoline, you contribute to carbon emissions.

You leave your carbon footprint.

You can lessen your carbon footprint by walking, riding a bike, taking a cold shower, or planting a tree.

But can each one of us be relied upon to make decisions that affect the Earth's carbon-dioxide concentration in the social interest?

Can governments change the incentives we face so that our self-interested choices are also in the social interest?

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Two Big Economic Questions

lockdown led to a decline in the economy

The Covid Pandemic

Covid-19 spreads through close contact and social interaction.

With no social isolation, one infected person infects more than one other person, each infected person infects more than one other, leading to an exponential spread of the disease?

A person who socially isolates avoids infection, but also avoids infecting others. People choose their degree of social isolation in their self-interest.

Do individual choices serve the social interest?

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Economic Way of Thinking

Six key ideas define the economic way of thinking:

- A choice is a *tradeoff*.
- People make *rational choices* by comparing *benefits and costs*.
- *Benefit* is *what you gain* from something.
- *Cost* is what you *must give up* to get something.
- Most choices are *"how-much"* choices made at the *margin*.
- Choices respond to *incentives*.

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Economic Way of Thinking

A Choice Is a Tradeoff

If you increase one production you must decrease another due to scarce resources which must be used productively and efficiently

The *economic way of thinking* places *scarcity* and its implication, *choice*, at center stage.

You can think about every choice as a **tradeoff**—an exchange—giving up one thing to get something else.

On Saturday night, will you study or have fun?

You can't study and have fun at the same time, so *y*ou must make a choice.

Whatever you choose, you could have chosen something else. Your choice is a tradeoff.

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Economic Way of Thinking

Making a Rational Choice

A **rational choice** is one that compares costs and benefits and achieves the greatest benefit over cost for the person making the choice.

Only the wants of the person making a choice are relevant to determine its rationality.

The idea of rational choice provides an answer to the first question: *What* goods and services will be produced and in what quantities?

The answer is: Those that people rationally choose to buy!

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The Economic Way of Thinking

How do people choose rationally?

The answers turn on benefits and costs.

Benefit: What you Gain

The **benefit** of something is the gain or pleasure that it brings and is determined by preferences

Preferences are what a person likes and dislikes and the intensity of those feelings.

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The Economic Way of Thinking

Cost: What you *Must* Give Up

The **opportunity cost** of something is the highest-valued alternative that *must* be given up to get it.

next best choice you are sacrificing

What is your opportunity cost of going to a live concert?

Opportunity cost has two components:

1. The things you can't afford to buy if you purchase the concert ticket.

2. The things you can't do with your time if you attend the concert.

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The Economic Way of Thinking

How Much? Choosing at the Margin

You can allocate the next hour between studying and instant messaging your friends.

The choice is not all or nothing, but you must decide how many minutes to allocate to each activity.

To make this decision, you compare the benefit of a little bit more study time with its cost—you make your choice at the **margin**.

↓ extra/addition that presents tradeoffs

e.g. monthly budget - will be disturbed if something unusual happens

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The Economic Way of Thinking

To make a choice at the margin, you evaluate the consequences of making *incremental changes* in the use of your time.

The benefit from pursuing an incremental increase in an activity is its **marginal benefit**.

The opportunity cost of pursuing an incremental increase in an activity is its **marginal cost**.

If the marginal benefit from an incremental increase in an activity exceeds its marginal cost, your rational choice is to do more of that activity.

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The Economic Way of Thinking

Choices Respond to Incentives

A change in marginal cost or a change in marginal benefit changes the incentives that we face and leads us to change our choice.

The central idea of economics is that we can predict how choices will change by looking at changes in incentives.

Incentives are also the key to reconciling self-interest and the social interest.

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Economics: A Social Science and Policy Tool

Economist as Social Scientist

Economists distinguish between two types of statement:

- Positive statements → what is
- Normative statements → what ought to be

A *positive* statement can be tested by checking it against facts. → factual statements

A *normative* statement expresses an opinion and cannot be tested. → opinionated statement
value-judgement

★ understand it by linking it to real-life situations

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Economics: A Social Science and Policy Tool

Unscrambling Cause and Effect

The task of economic science is to discover positive statements that are consistent with what we observe in the world and that enable us to understand how the economic world works.

researches who assess current/past data to forecast future in order to formulate policies

Economists create and test economic models.

An **economic model** is a description of some aspect of the economic world that includes only those features that are needed for the purpose at hand. e.g. Google Maps

similar to models used in other sciences as they attempt to study reality - providing general + testable statements

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to tackle a given

Economics: A Social Science and Policy Tool

A model is tested by comparing its predictions with the facts.

But testing an economic model is difficult, so economists also use:

- Natural experiments
- Statistical investigations
- Economic experiments

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Economics: A Social Science and Policy Tool

Economist as Policy Adviser

Economics is a toolkit for advising governments and businesses and for making personal decisions.

All the policy questions on which economists provide advice involve a blend of the positive and the normative.

Economics can't help with the normative part—the goal.

But for a given goal, economics provides a method of evaluating alternative solutions—comparing marginal benefits and marginal costs.

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Economists in the Economy

What are the jobs available to an economics major?

Is the number of economics jobs expected to grow or shrink?

How much do economics graduates earn?

What are the skills needed for an economics job?

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Economists in the Economy

Jobs for an Economics Major

A major in economics opens the door to the pursuit of a masters or PhD and a career as an *economist*.

The work of economists varies enormously, but it includes:

Collecting and analyzing data on the production and use of resources, goods, and services;

Predicting future trends; and

Studying ways of using resources more efficiently.

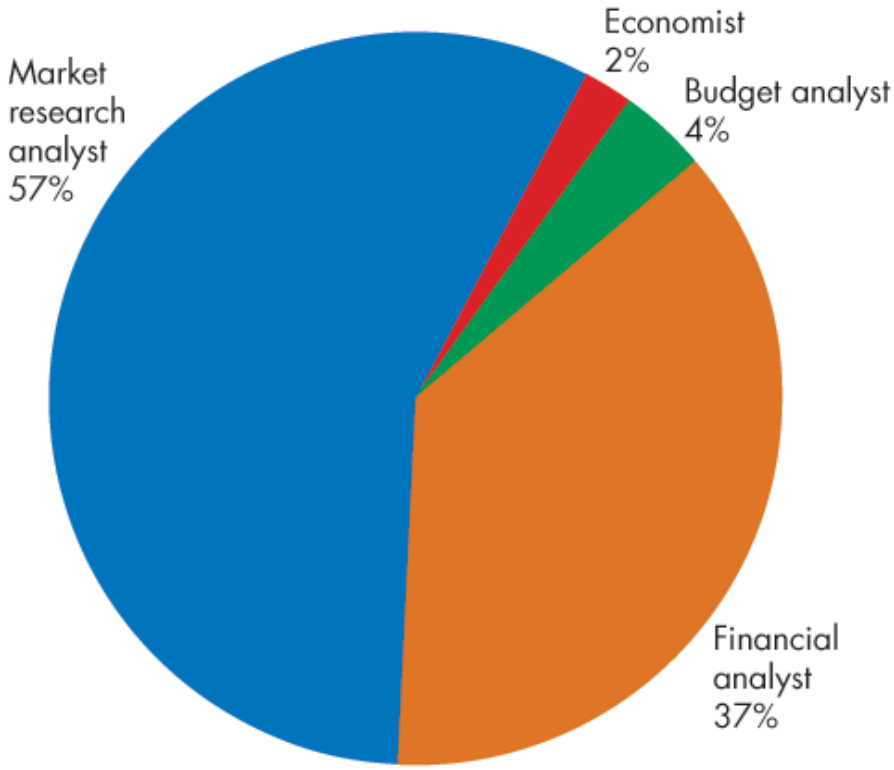
Economists work in private firms, government, and international organizations.

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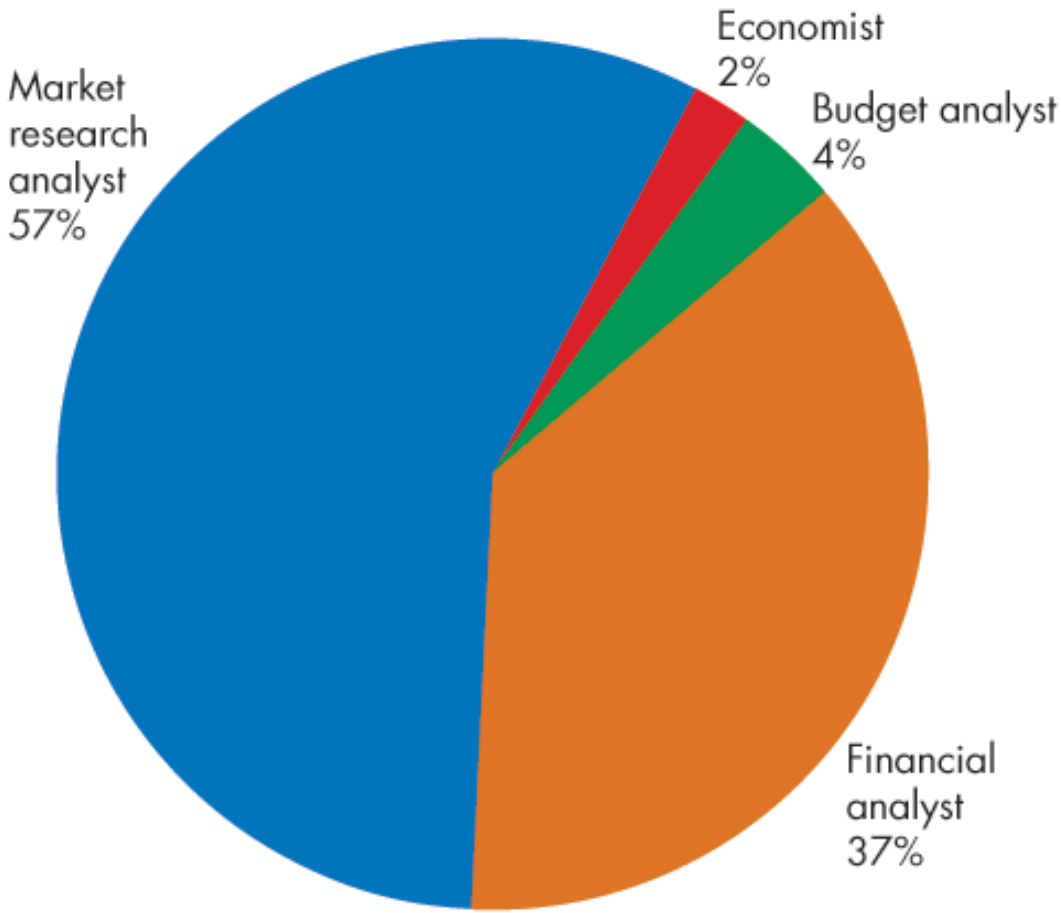
Economists in the Economy

Economics majors also work as market research analysts, financial analysts, and budget analysts.

Figure 1.3 shows the the relative number of jobs for economists and analysts that use economic ideas and tools.



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Economists in the Economy

Will Jobs for Economists Grow?

The BLS forecasts that from 2019 to 2029 jobs for:

1. Economists with a PhD will grow by 14 percent.
2. Budget analysts will grow by 3 percent.
3. Financial analysts will grow by 5 percent.
4. Market research analysts will grow by 18 percent.

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Economists in the Economy

Earnings of Economics Majors

Earnings depend on the job and the level of qualifications.

Web source payscale.com reports economists earn from $56,500 to $139,600, with a median of $105,100 a year.

An economics major with a PhD, who gets a job as an economist, earns about $150,000 a year by mid-career.

Pay in “analyst” jobs averages $55,000 a year for market research and $82,000 a year for financial analysts.

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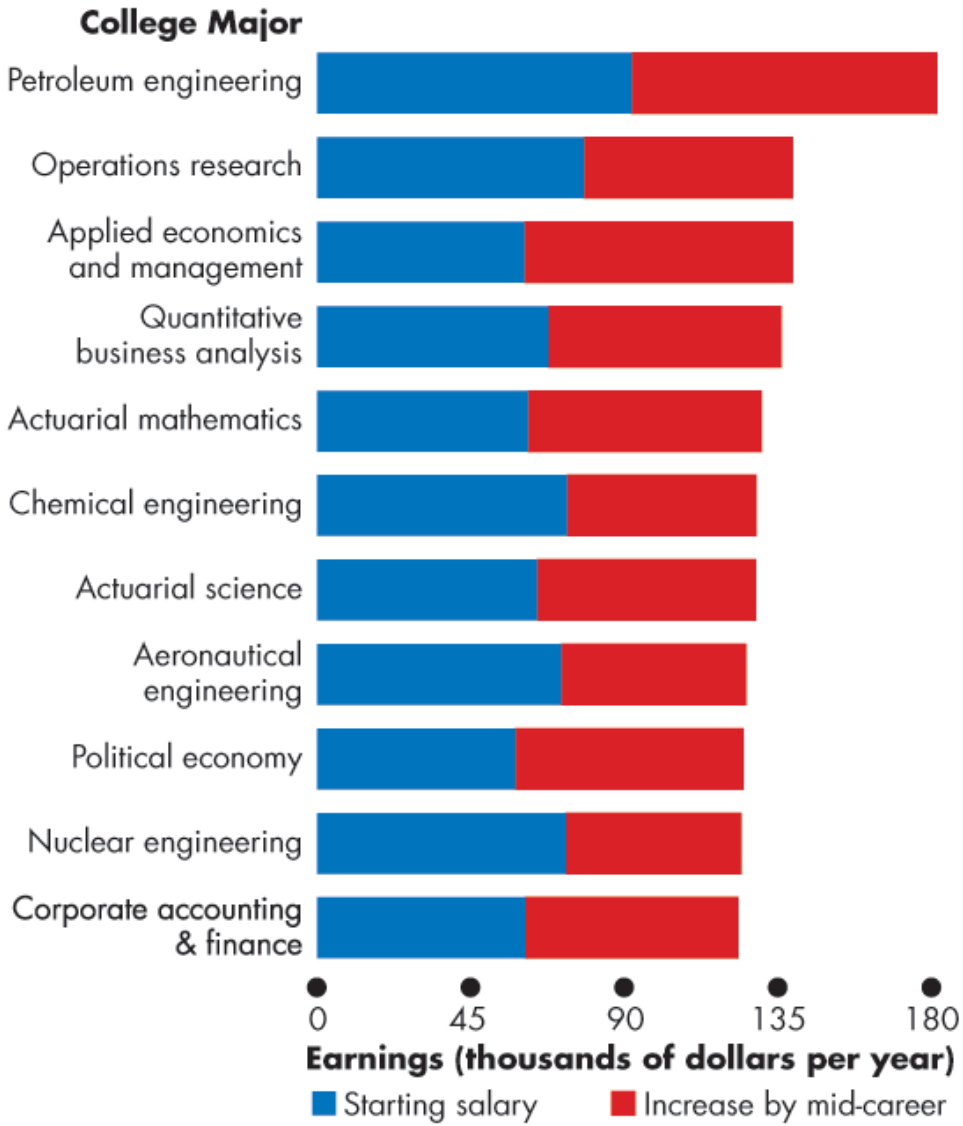
Economists in the Economy

Figure 1.4 shows starting salaries of college graduates

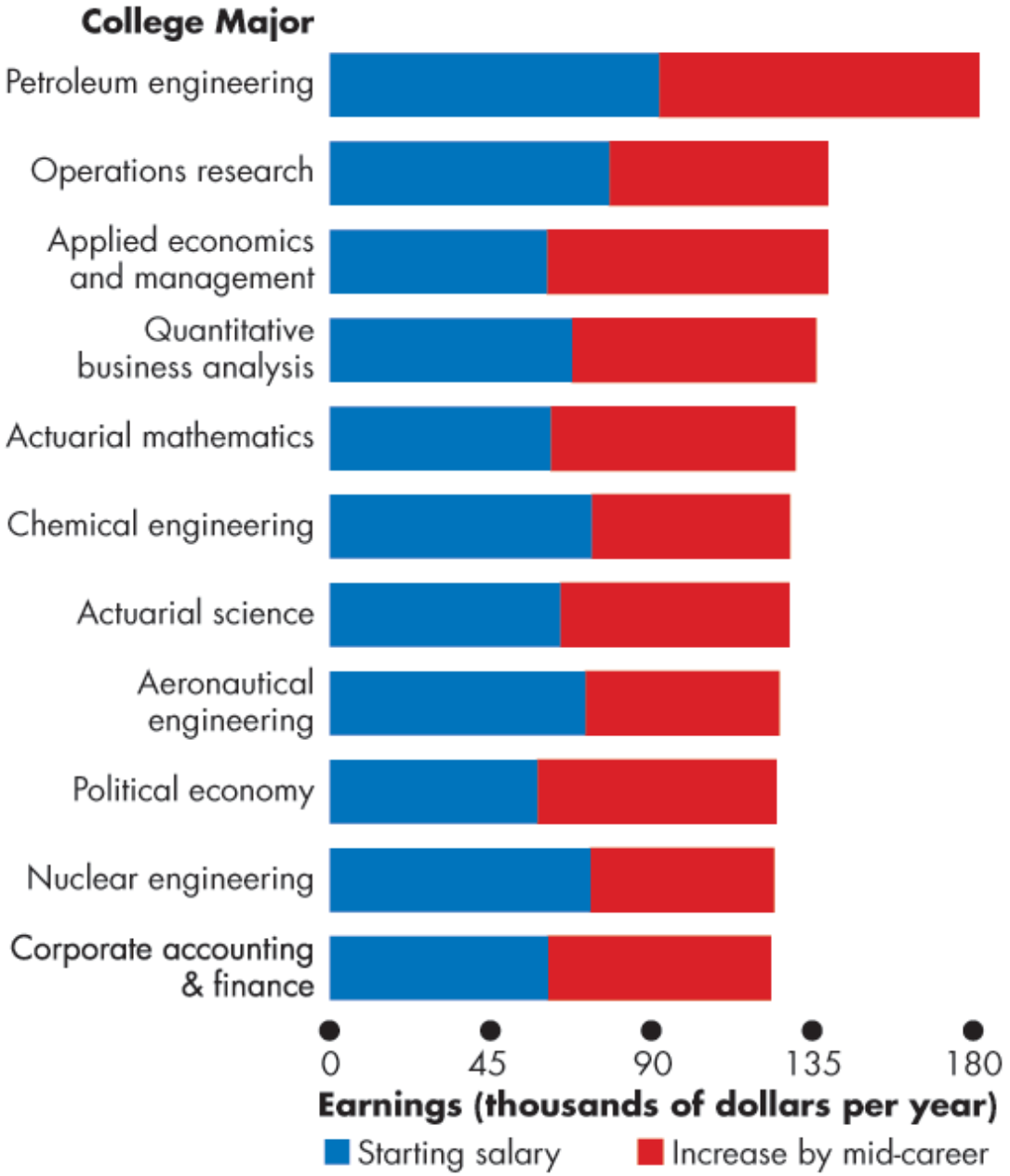
and mid-career salaries.

Economics majors are not the highest earners.

But at mid-career, they earn $140,000 a year, more than most other majors.



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Economists in the Economy

Skills Needed for Economics Jobs

Employers look for five skills:

1. Critical-thinking skills.
2. Analytical skills
3. Math skills
4. Writing skills
5. Oral communication skills

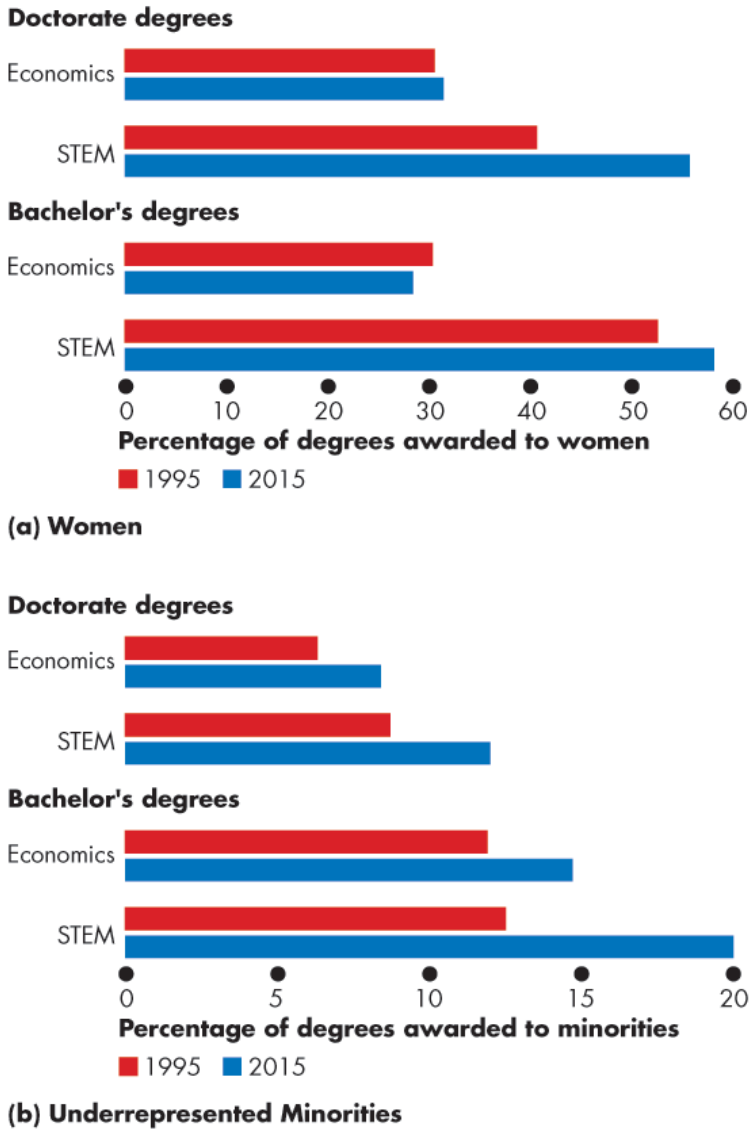
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Economists in the Economy

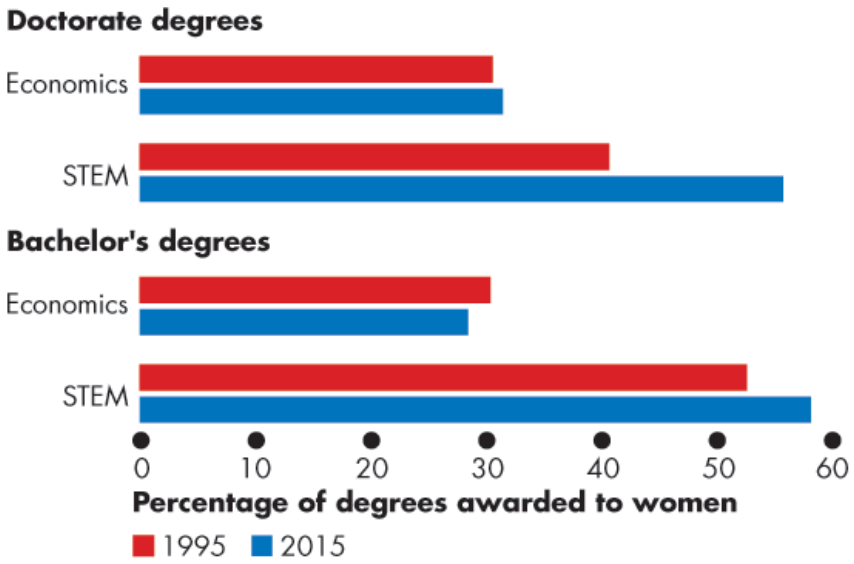
Diversity Challenge for Economics Profession

Compare the change from 1995 to 2015 in the percentages of women and minorities with economics degrees and STEM degrees.

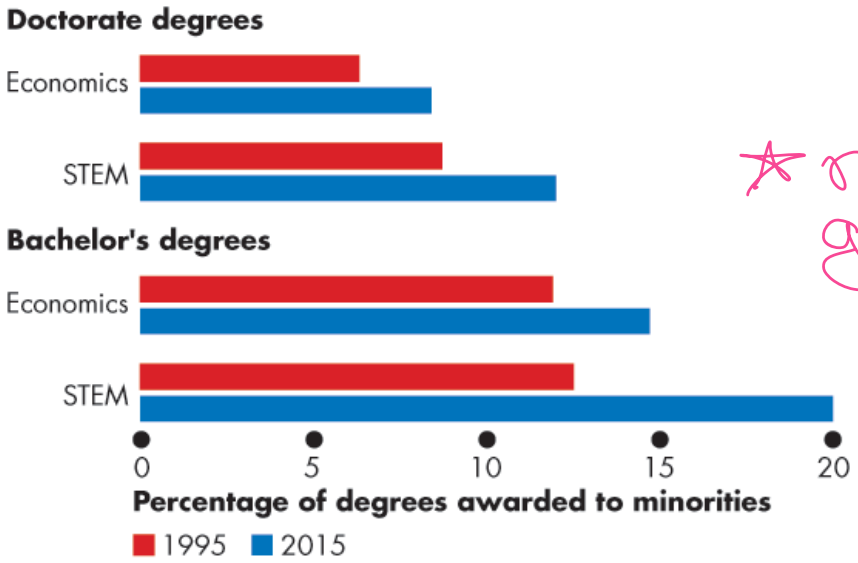
Economics profession needs to attract more women and minorities at all levels.



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(a) Women



(b) Underrepresented Minorities

★ refer to the graphs for understanding purposes

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APPENDIX

Graphs in Economics

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After studying this chapter, you will be able to:

- Make and interpret a scatter diagram
- Identify linear and nonlinear relationships and relationships that have a maximum and a minimum
- Define and calculate the slope of a line
- Graph relationships among more than two variables

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Graphing Data

A graph reveals a relationship.

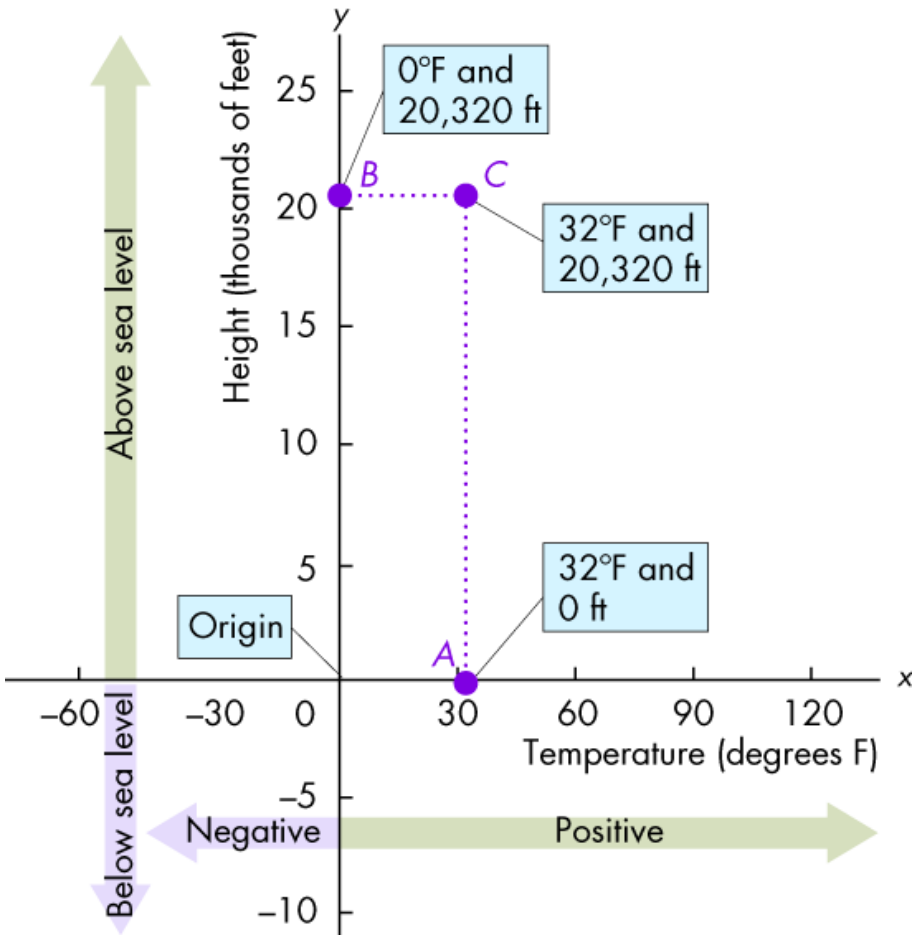
A graph represents “quantity” as a distance.

A two-variable graph uses two perpendicular scale lines.

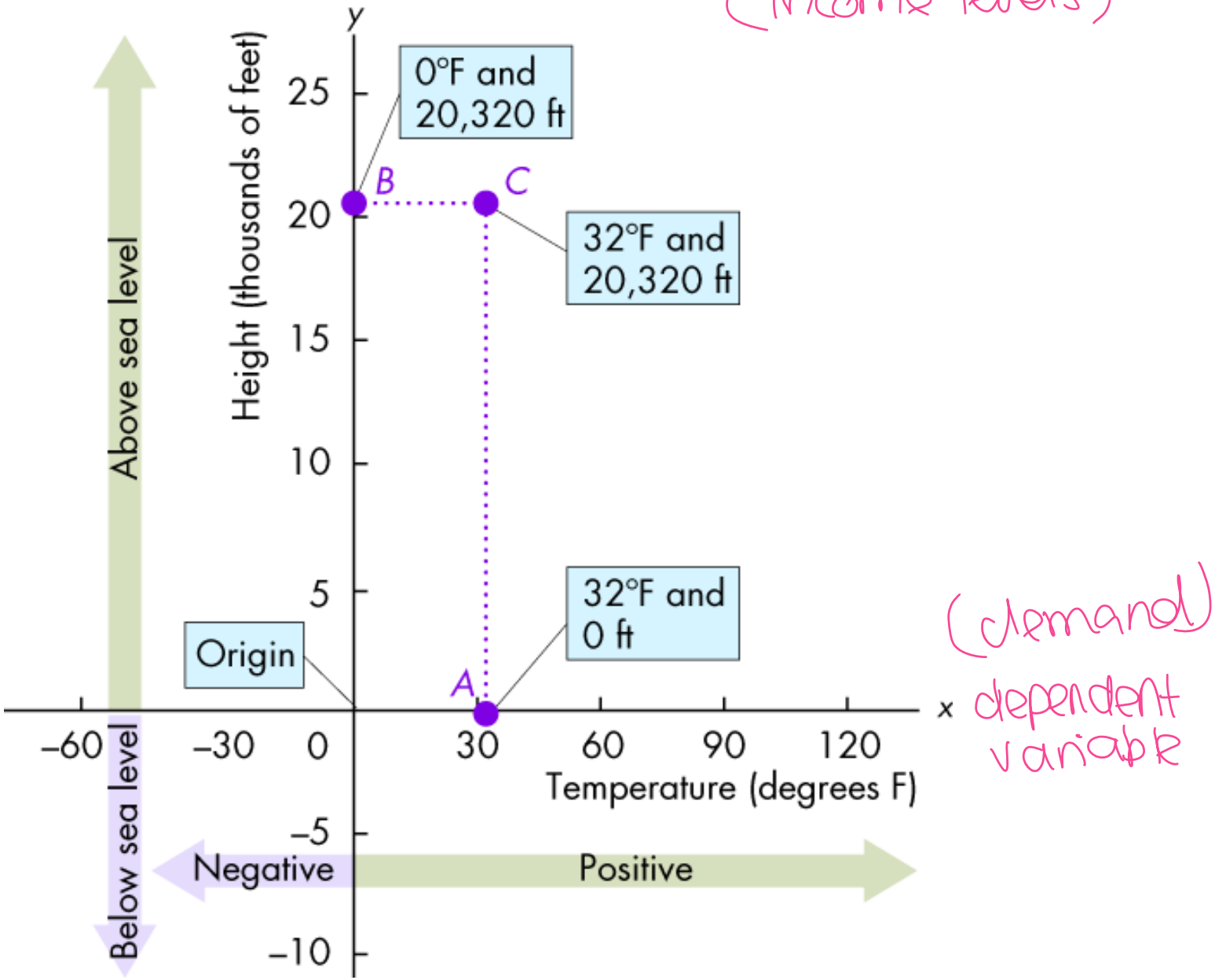
The horizontal line is the *x*-axis.

The vertical line is the *y*-axis.

The zero point in common to both axes is the *origin*.



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independent var (price)
(income levels)

(demand)
dependent variable

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Graphing Data

Economists measure variables that describe *what*, *how*, and *for whom* goods and services are produced.

These variables are quantities produced and prices.

Figure A1.2 shows two examples of economic graphs.

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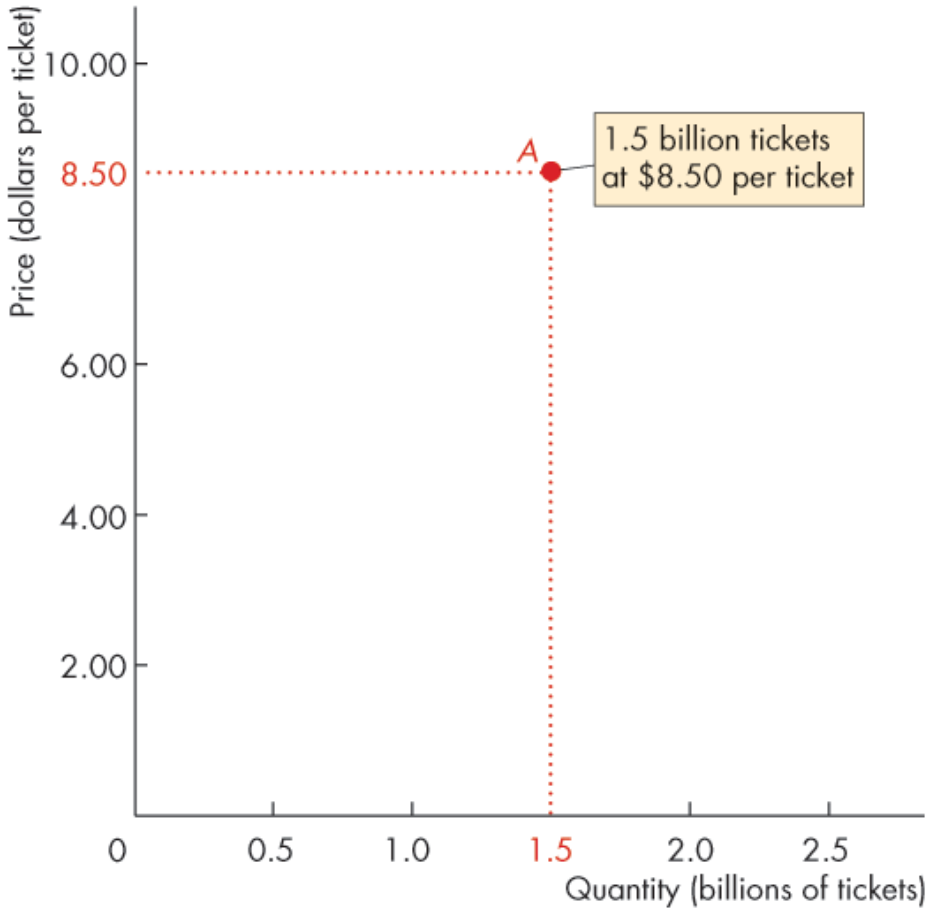
Graphing Data

Figure A1.2 shows how to make an economics graph.

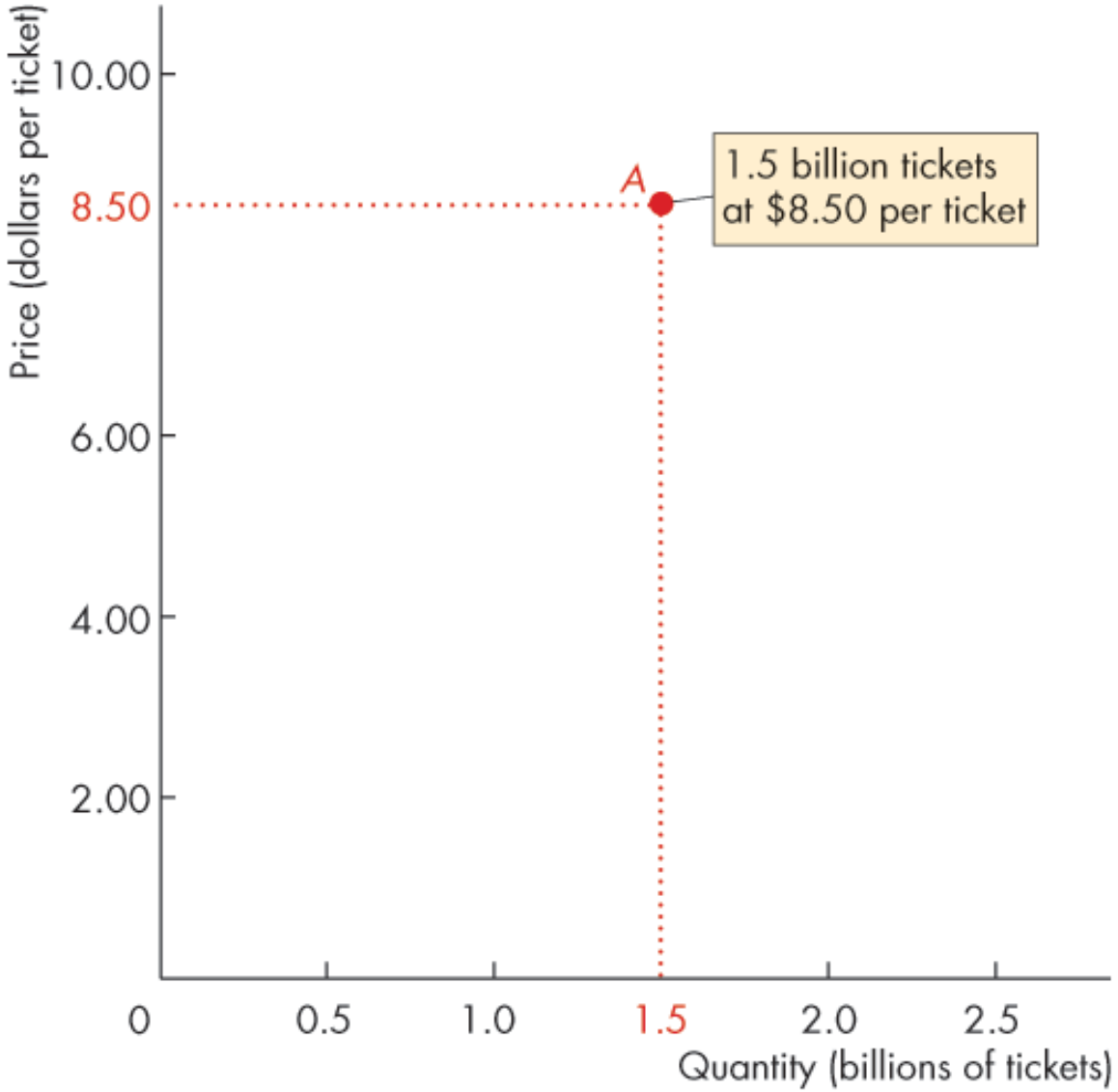
Point *A* tells us the quantity of tickets bought in 2020 and the average price of a ticket.

You can "read" this graph as telling you that in 2020:

1.5 billion movie tickets were bought at a price of $8.50 a ticket.



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Graphing Data

Scatter Diagrams

A **scatter diagram** plots the value of one variable against the value of another variable for a number of different values of each variable.

A scatter diagram reveals whether a relationship exists between the two variables.

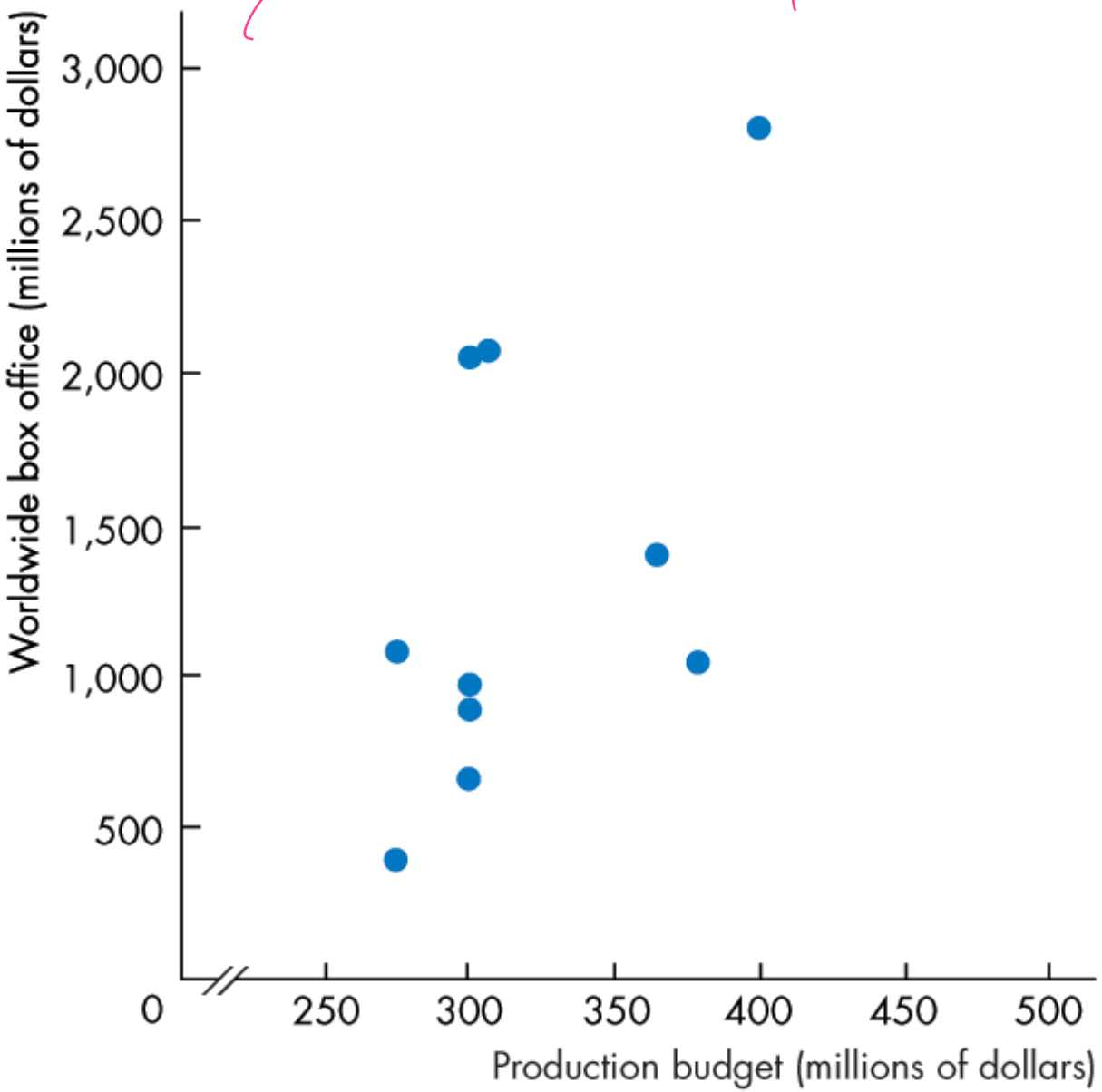
Figure A1.3 shows the production budget for ten popular movies and their worldwide box office revenues.

The table gives the data and the graph describes the relationship between each movie's production budget and its box office revenue.

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Graphing Data

Movie	Production budget (millions of dollars)	Worldwide box office (millions of dollars)
Avengers: Endgame	400	2,798
Pirates of the Caribbean: On Stranger Tides	379	1,046
Avengers: Age of Ultron	365	1,395
Star Wars Ep. VII: The Force Awakens	306	2,065
Pirates of the Caribbean: At World's End	300	961
Spectre	300	880
Avengers: Infinity War	300	2,045
Justice League	300	656
Star Wars: The Rise of Skywalker	275	1,073
Solo: A Star Wars Story	275	393

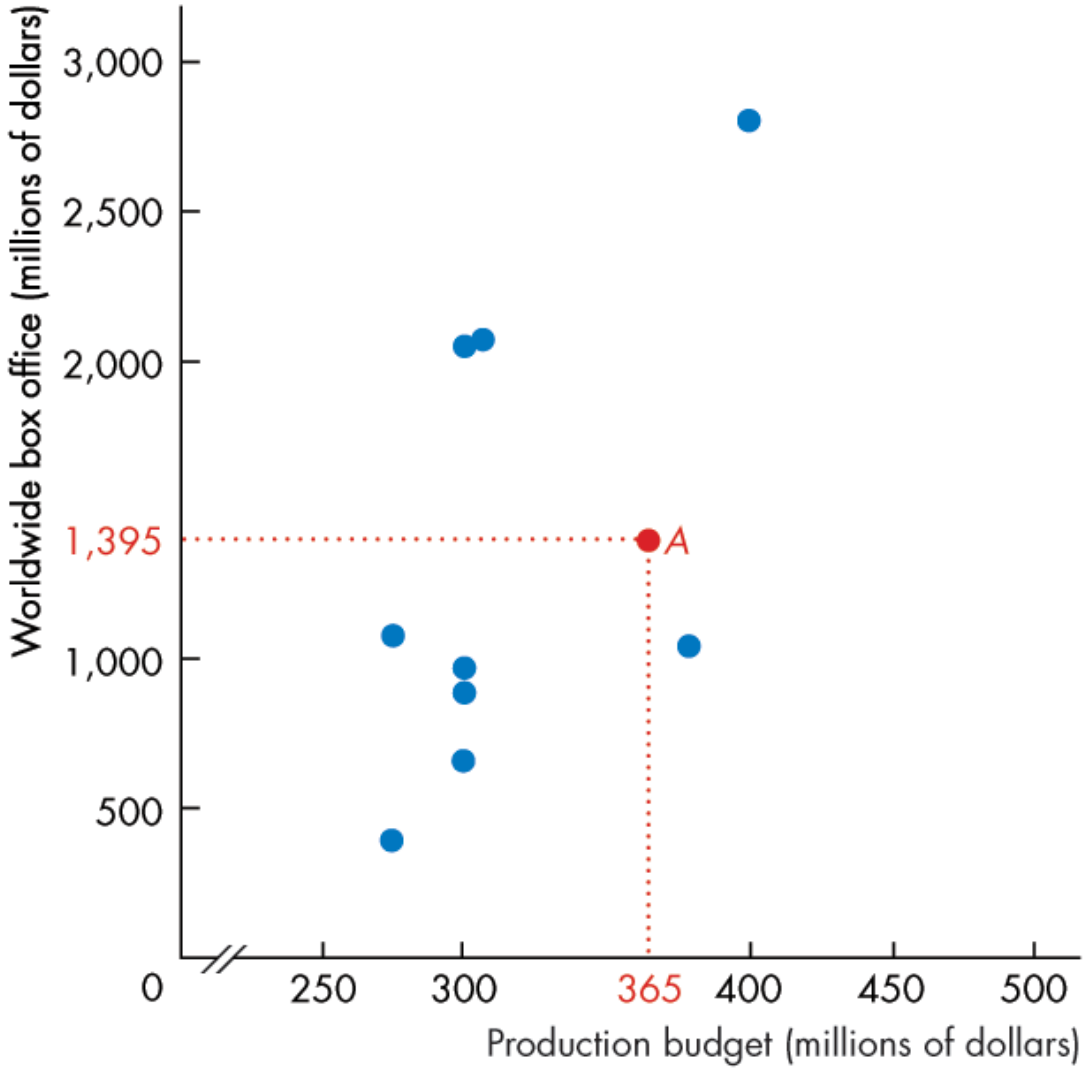


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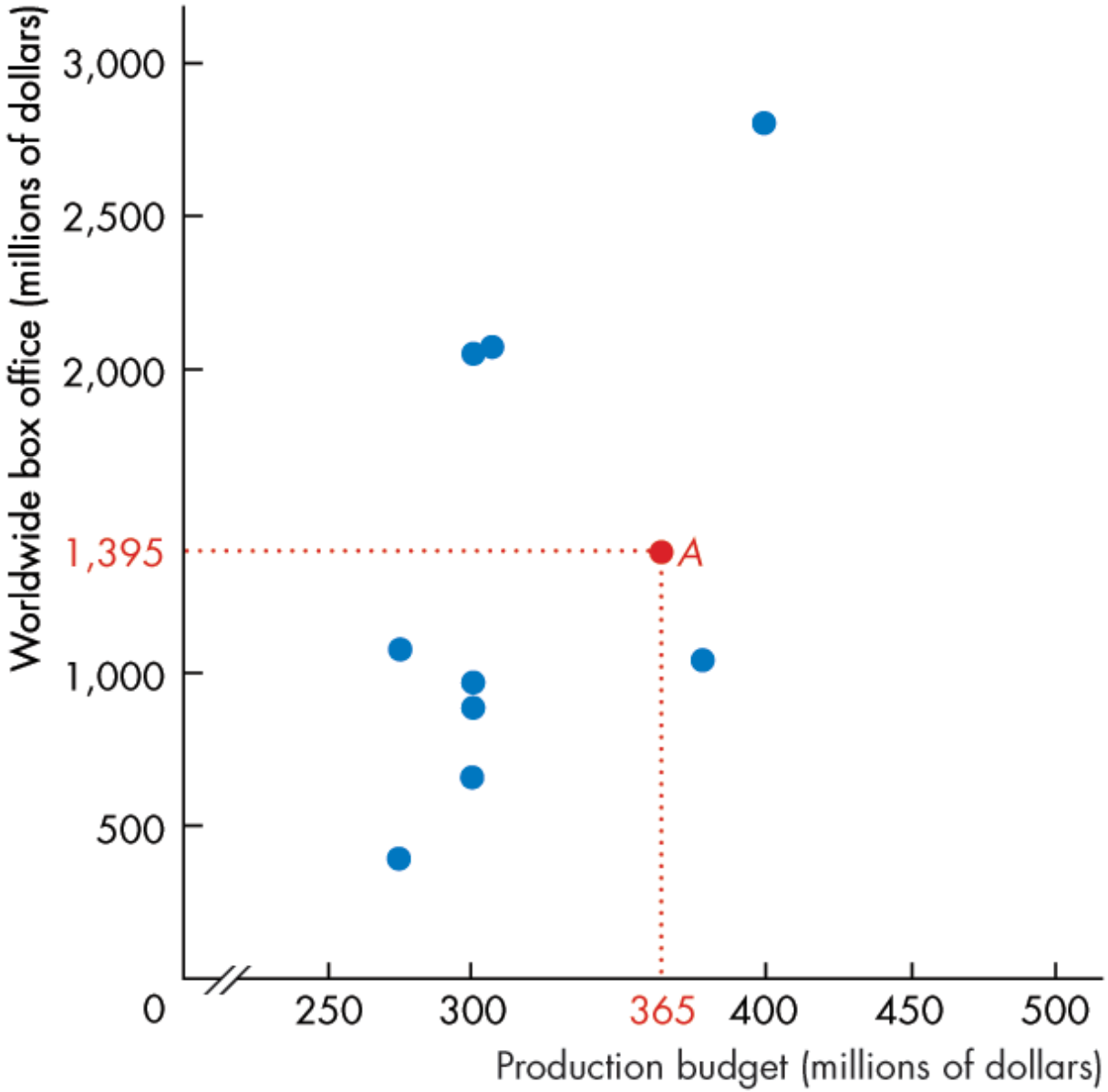
Graphing Data

Point *A* tells us that *Avengers: Age of Ultron* cost $365 million to produce and brought in $1,395 million at the box office.

The pattern of the points reveal that there is no clear tendency for a larger production budget to bring a greater box-office revenue.



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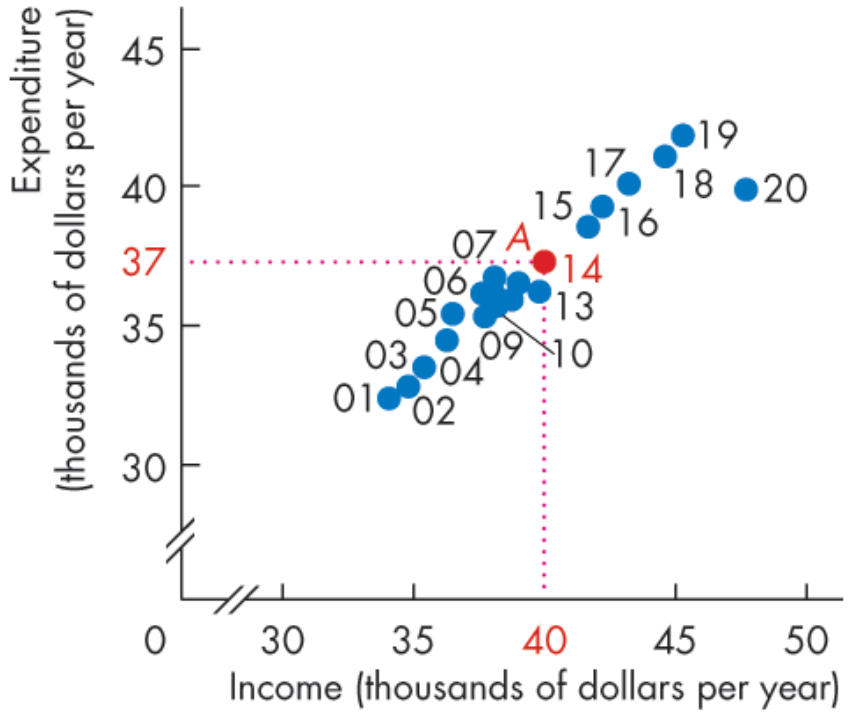
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Graphing Data

Figure A1.4(a) is a scatter diagram of income and expenditure, on average, from 2001 to 2020.

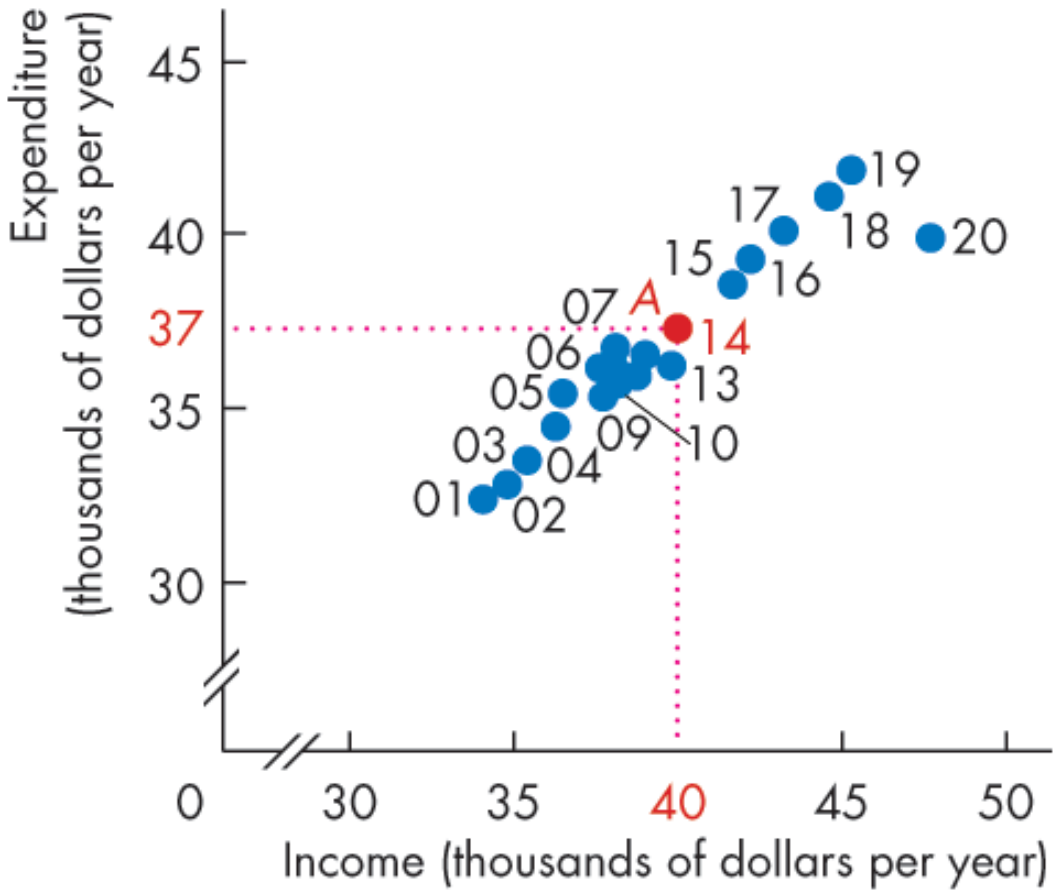
Point *A* shows that in 2014, income was $40,000 and expenditure was $37,000.

The graph shows that as income increases, so does expenditure, and that the relationship is a close one.



(a) Income and expenditure

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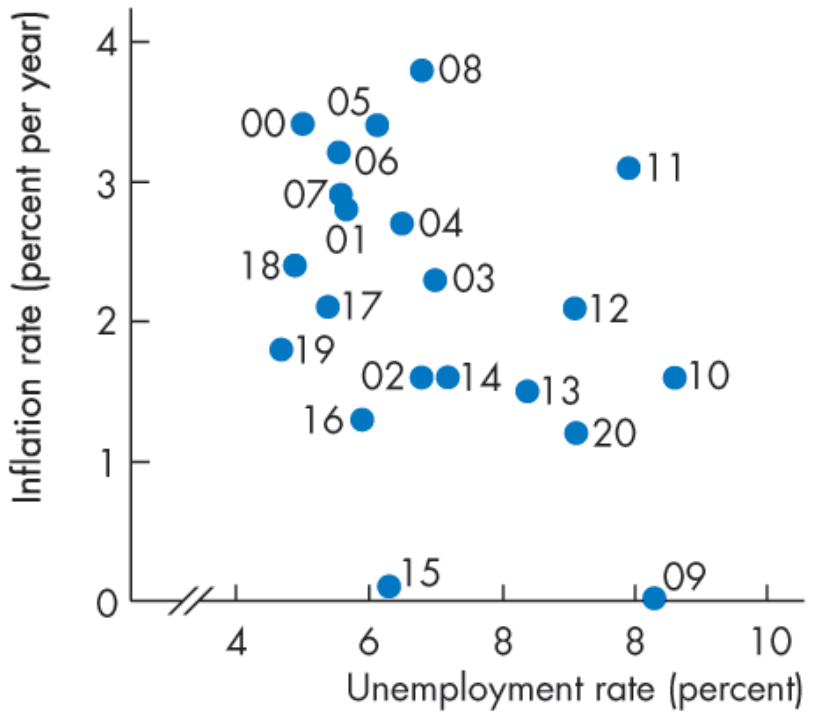
(a) Income and expenditure

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Graphing Data

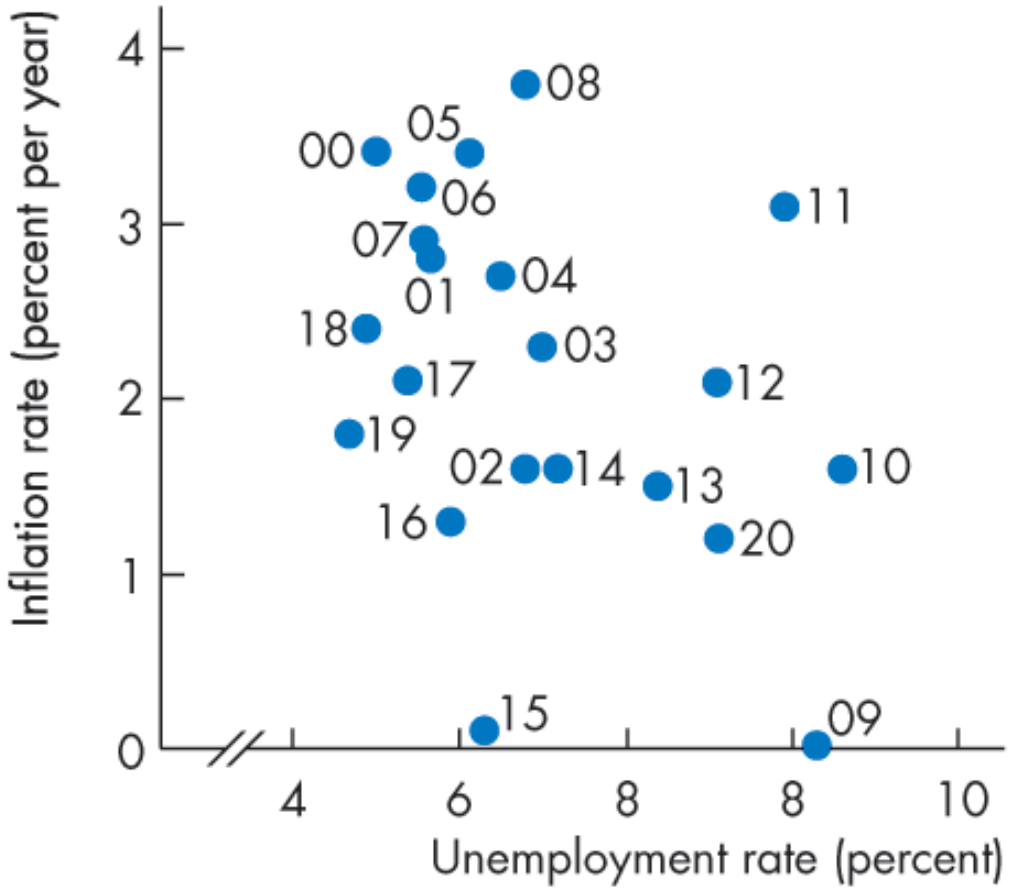
Figure A1.4(b) is a scatter diagram of inflation and unemployment in the United States from 2001 to 2020.

The points show a weak relationship between the two variables.



(b) Unemployment and inflation

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(b) Unemployment and inflation

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Graphs used in Economic Models

Graphs are used in economic models to show the relationship between variables.

The patterns to look for in graphs are the four cases in which

- Variables move in the same direction.
- Variables move in opposite directions.
- Variables have a maximum or a minimum.
- Variables are unrelated.

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Graphs Used in Economic Models

Variables That Move in the Same Direction

A relationship between two variables that move in the same direction is called a **positive relationship** or a **direct relationship**.

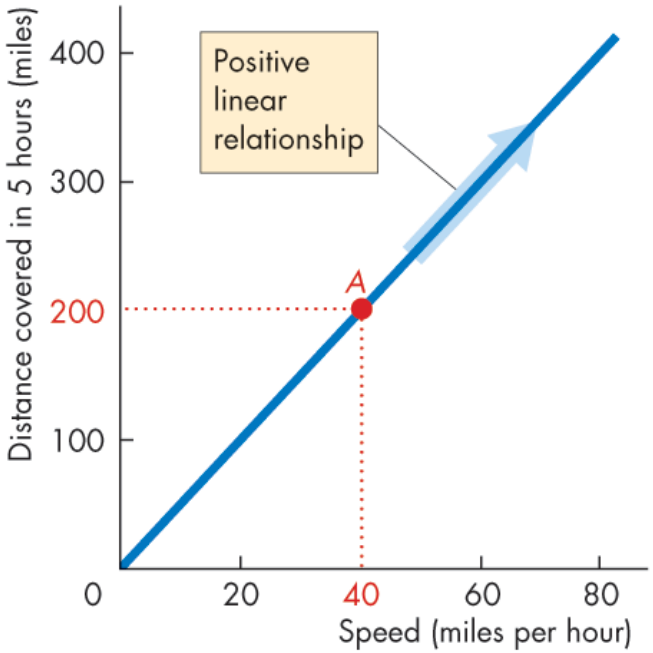
A line that slopes upward shows a positive relationship.

A relationship shown by a straight line is called a **linear relationship**.

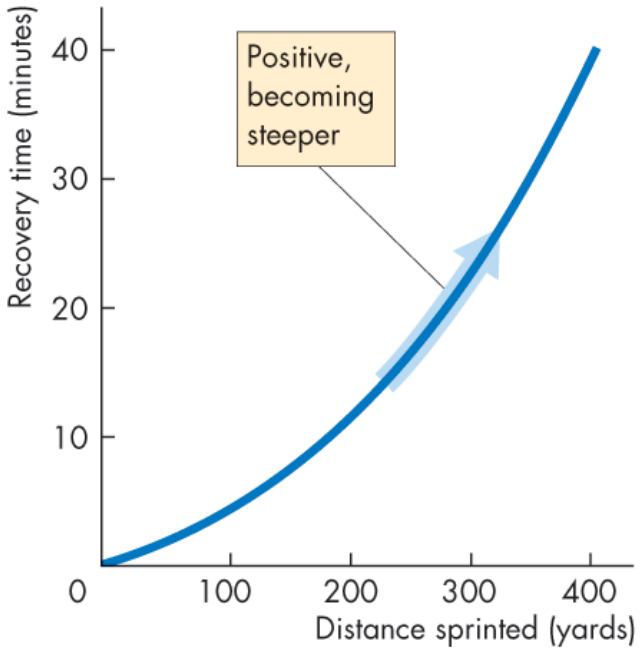
The three graphs on the next slide show positive relationships.

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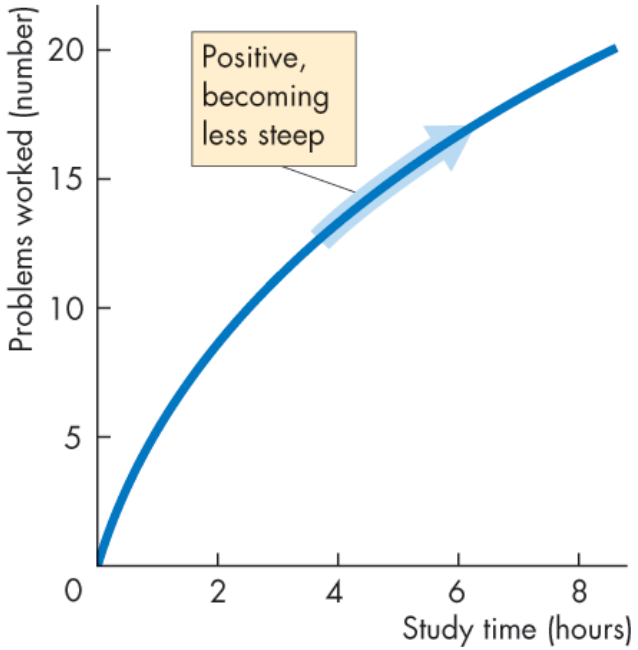
Graphs used in Economic Models



(a) Positive linear relationship



(b) Positive, becoming steeper



(c) Positive, becoming less steep

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Graphs used in Economic Models

Variables That Move in Opposite Directions

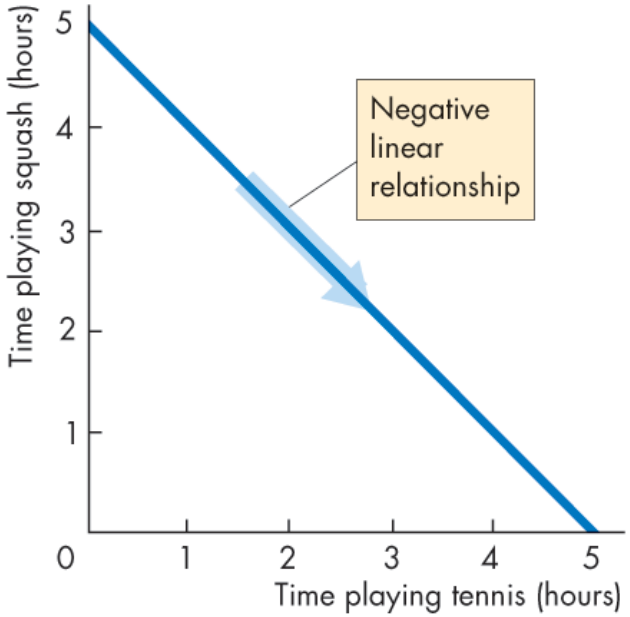
A relationship between two variables that move in opposite directions is called a **negative relationship** or an **inverse relationship**.

A line that slopes downward shows a negative relationship.

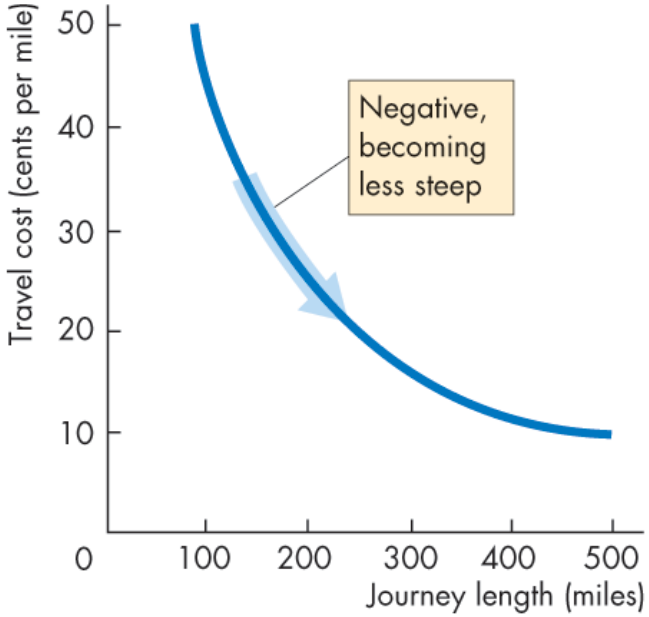
The three graphs on the next slide show negative relationships.

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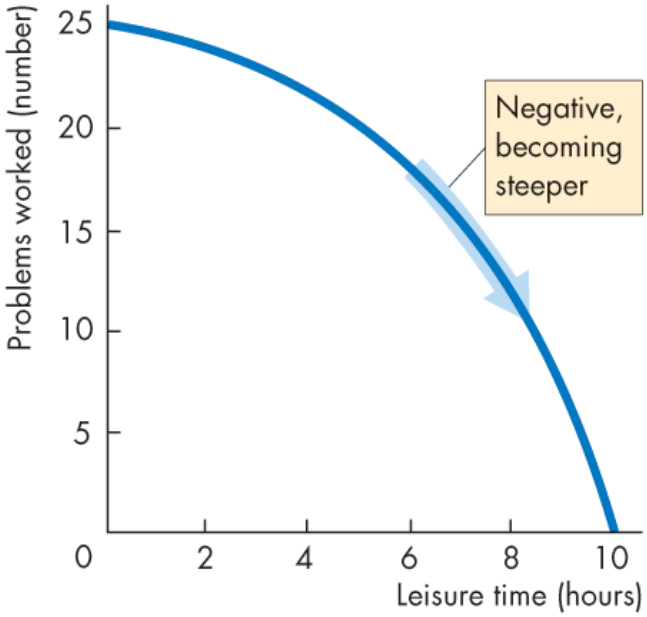
Graphs used in Economic Models



(a) Negative linear relationship



(b) Negative, becoming less steep



(c) Negative, becoming steeper

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Graphs used in Economic Models

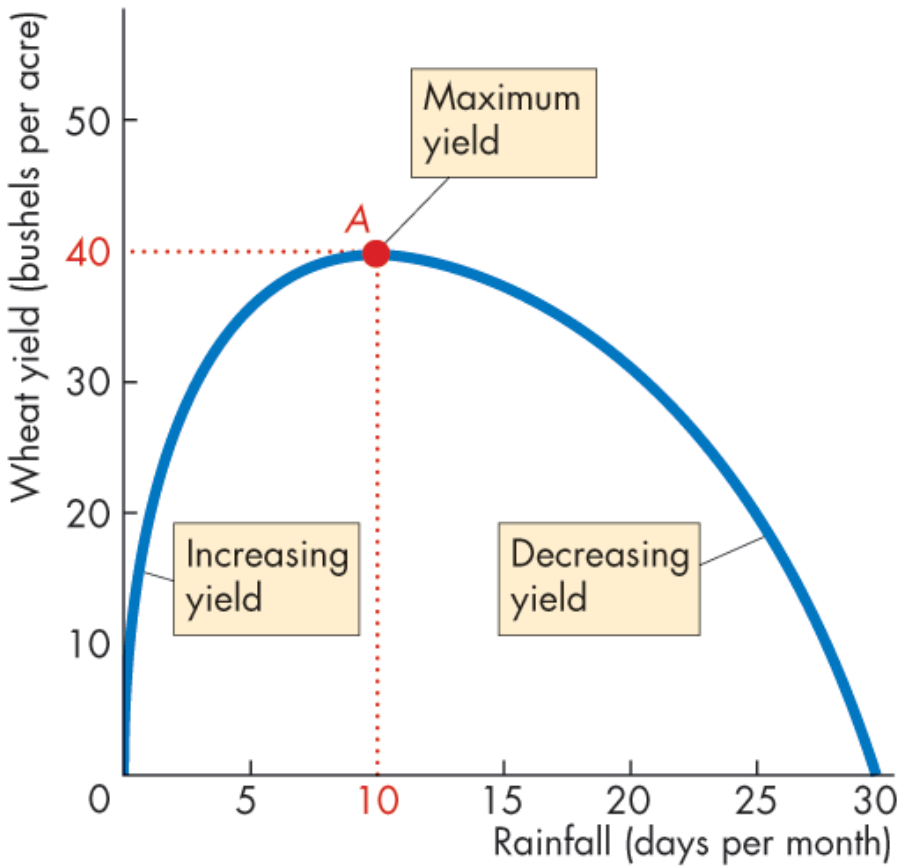
Variables That Have a Maximum or a Minimum

The two graphs on the next slide show relationships that have a maximum and a minimum.

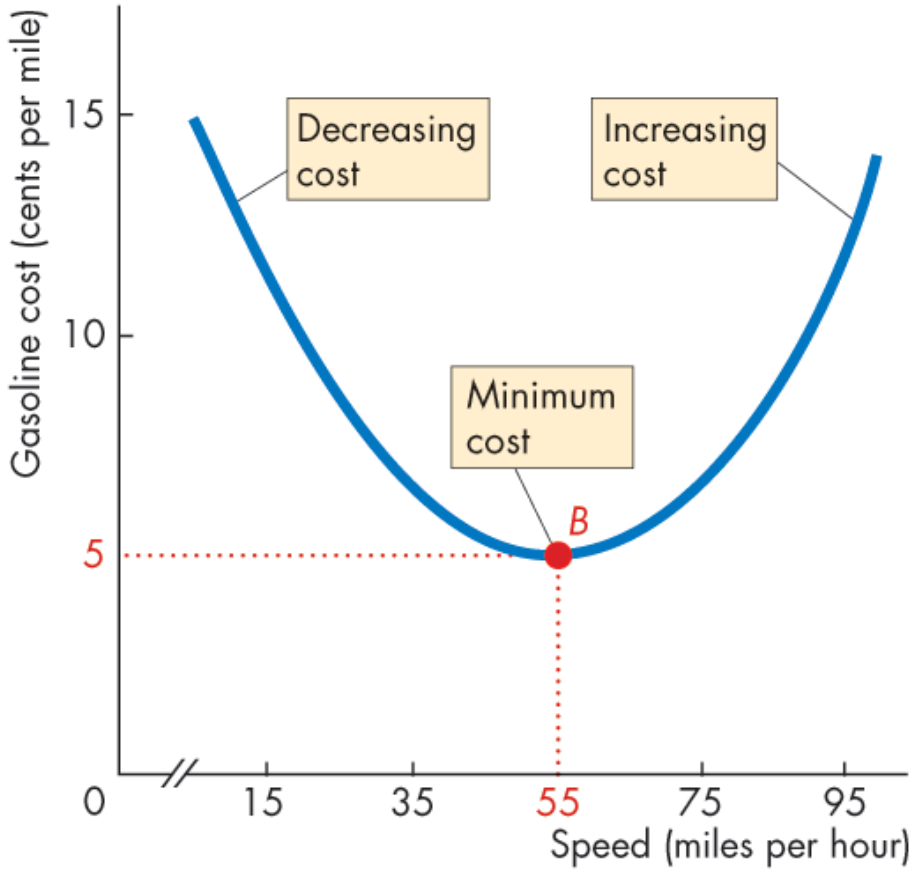
These relationships are positive over part of their range and negative over the other part.

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Graphs used in Economic Models



(a) Relationship with a maximum



(b) Relationship with a minimum

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Graphs used in Economic Models

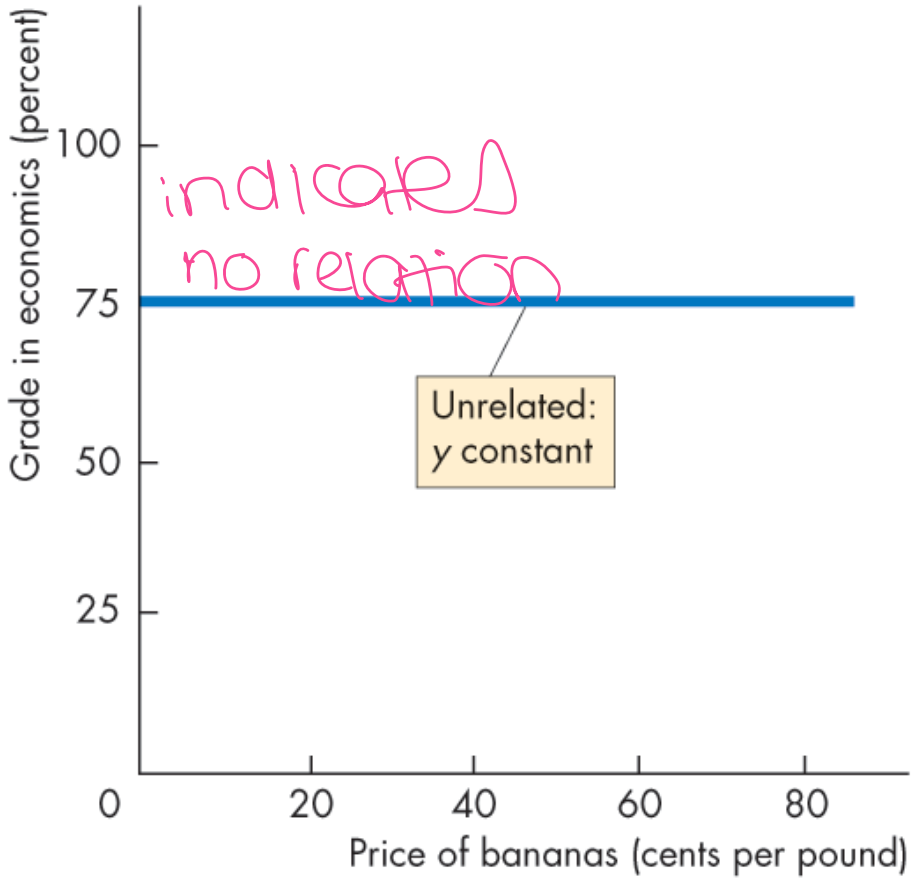
Variables That are Unrelated

Sometimes, we want to emphasize that two variables are unrelated.

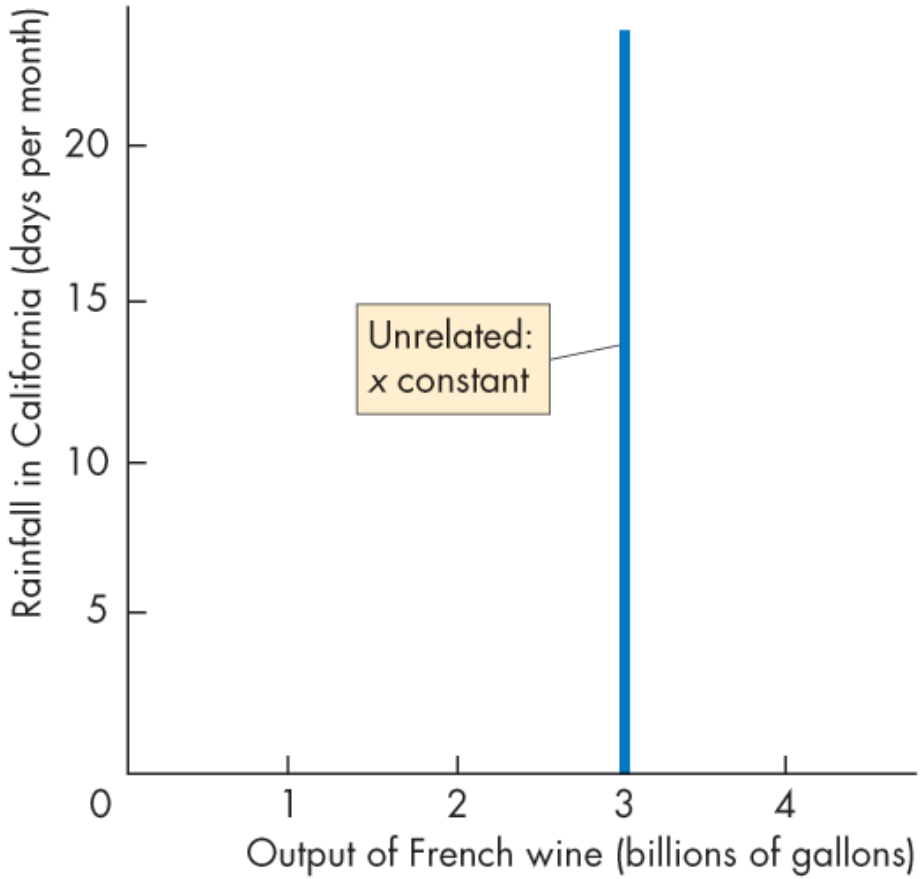
The two graphs on the next slide show examples of variables that are unrelated.

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Graphs used in Economic Models



(a) Unrelated: *y* constant



(b) Unrelated: *x* constant

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The Slope of a Relationship

The **slope** equals the change in the value of the variable measured on the *y*-axis divided by the change in the value of the variable measured on the *x*-axis.

We use the Greek letter (capital delta) to represent "change in."

So *y* means the change in the value of the variable measured on the *y*-axis and *x* means the change in the value of the variable measured on the *x*-axis.

Slope equals *y* *x*.

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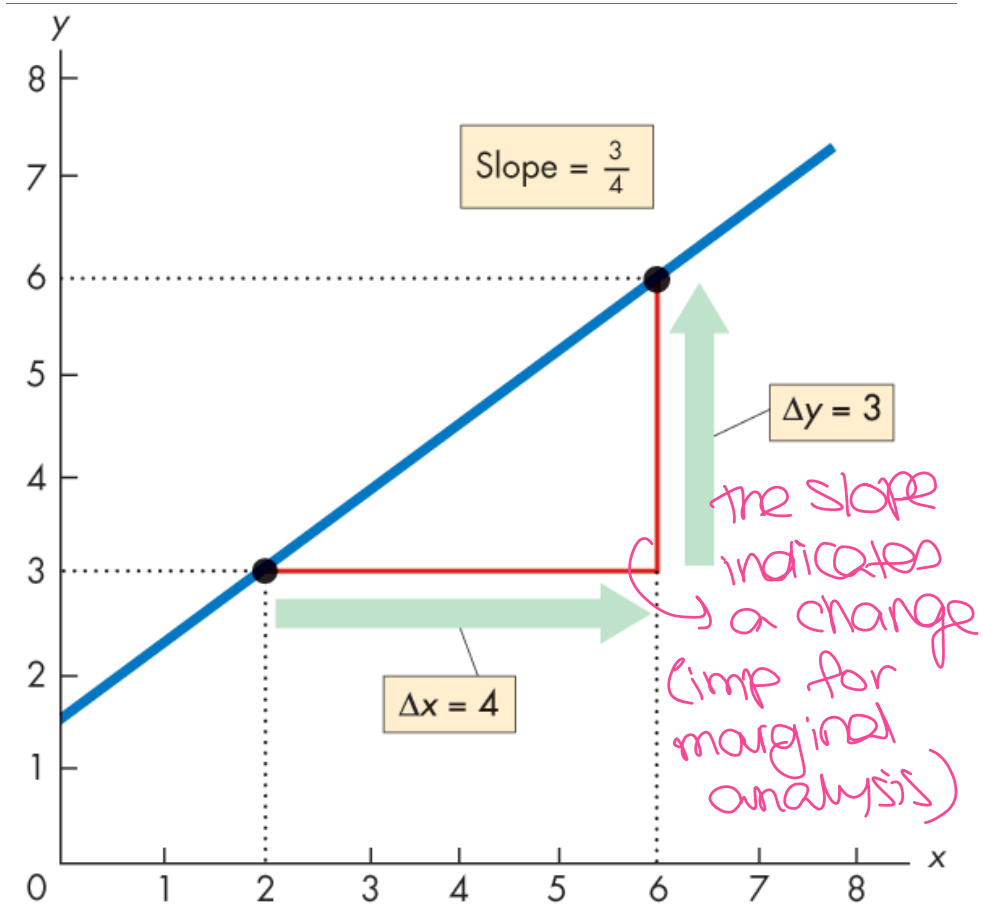
The Slope of a Relationship

Slope of a Straight Line

The slope of a straight line is constant.

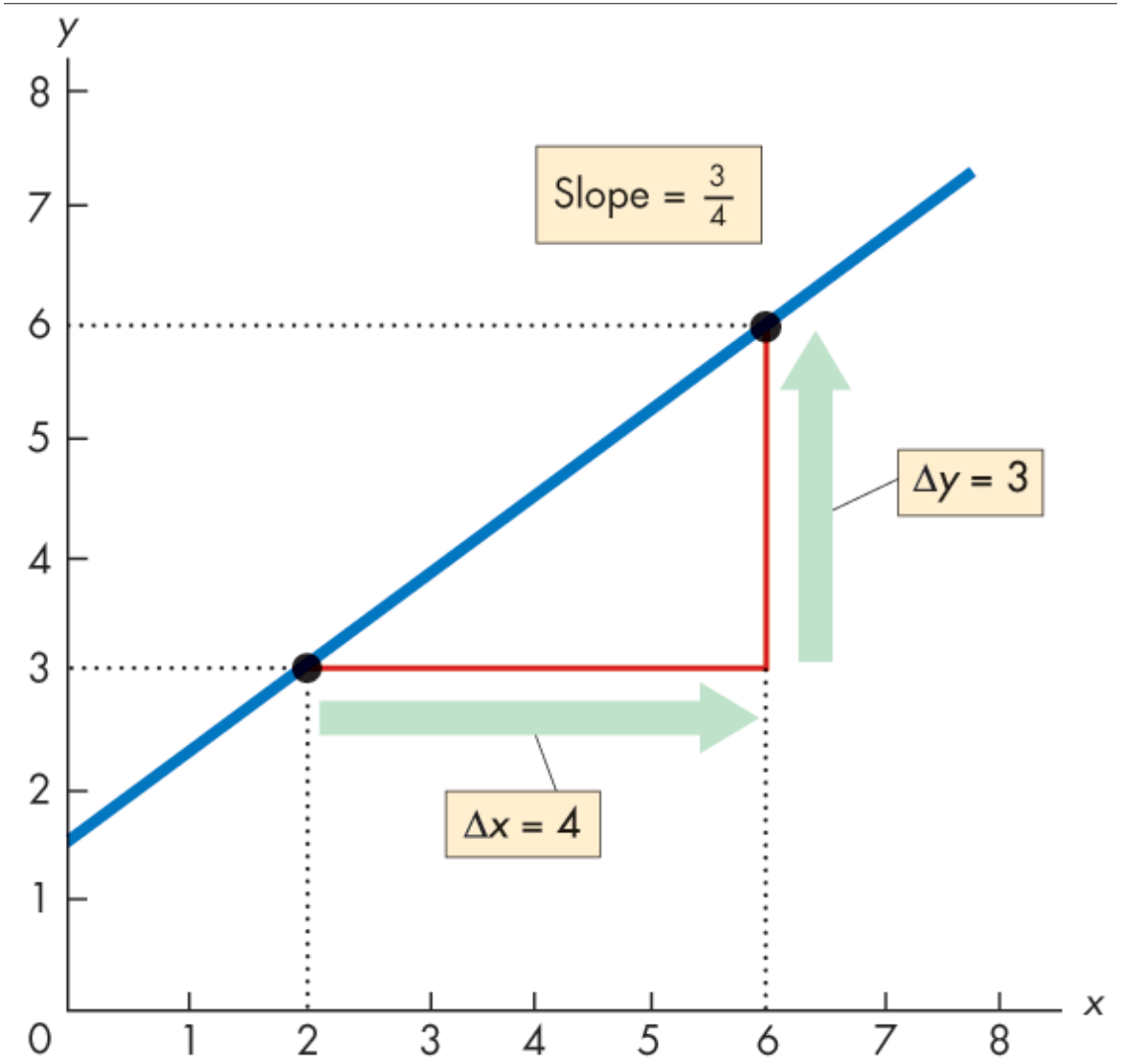
Graphically, the slope is calculated as the “rise” over the “run.”

The slope is positive if the line is upward sloping.



(a) Positive slope

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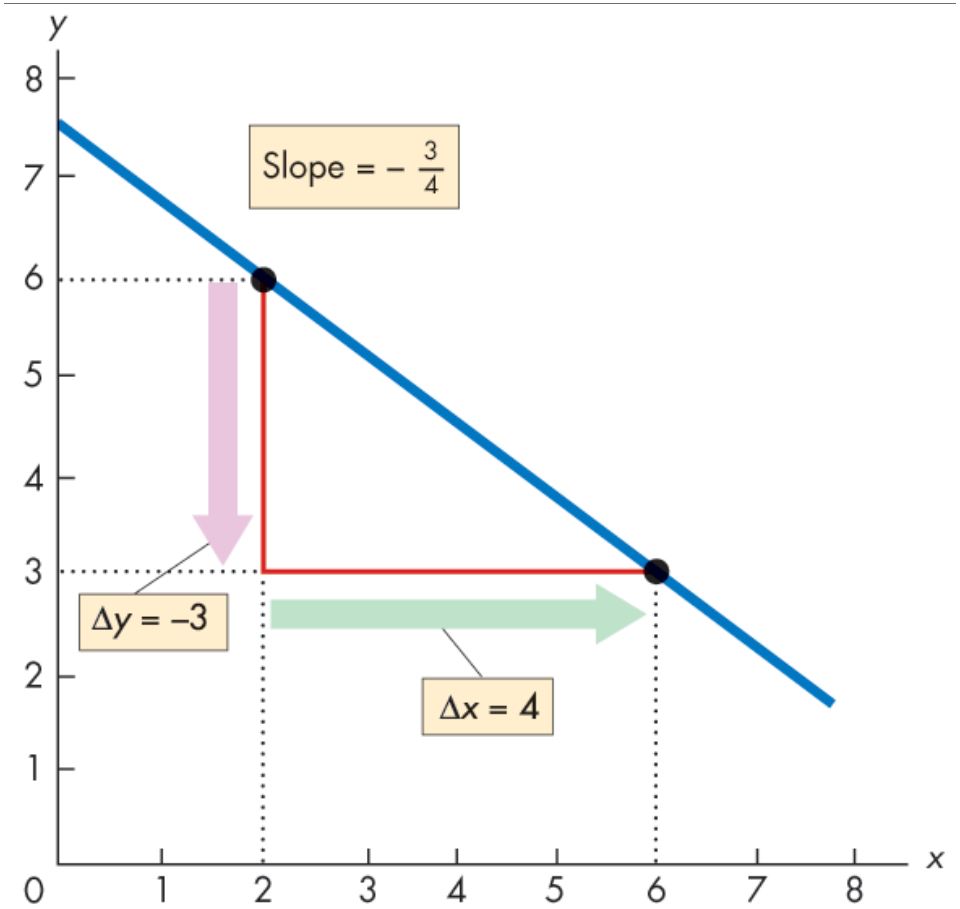


(a) Positive slope

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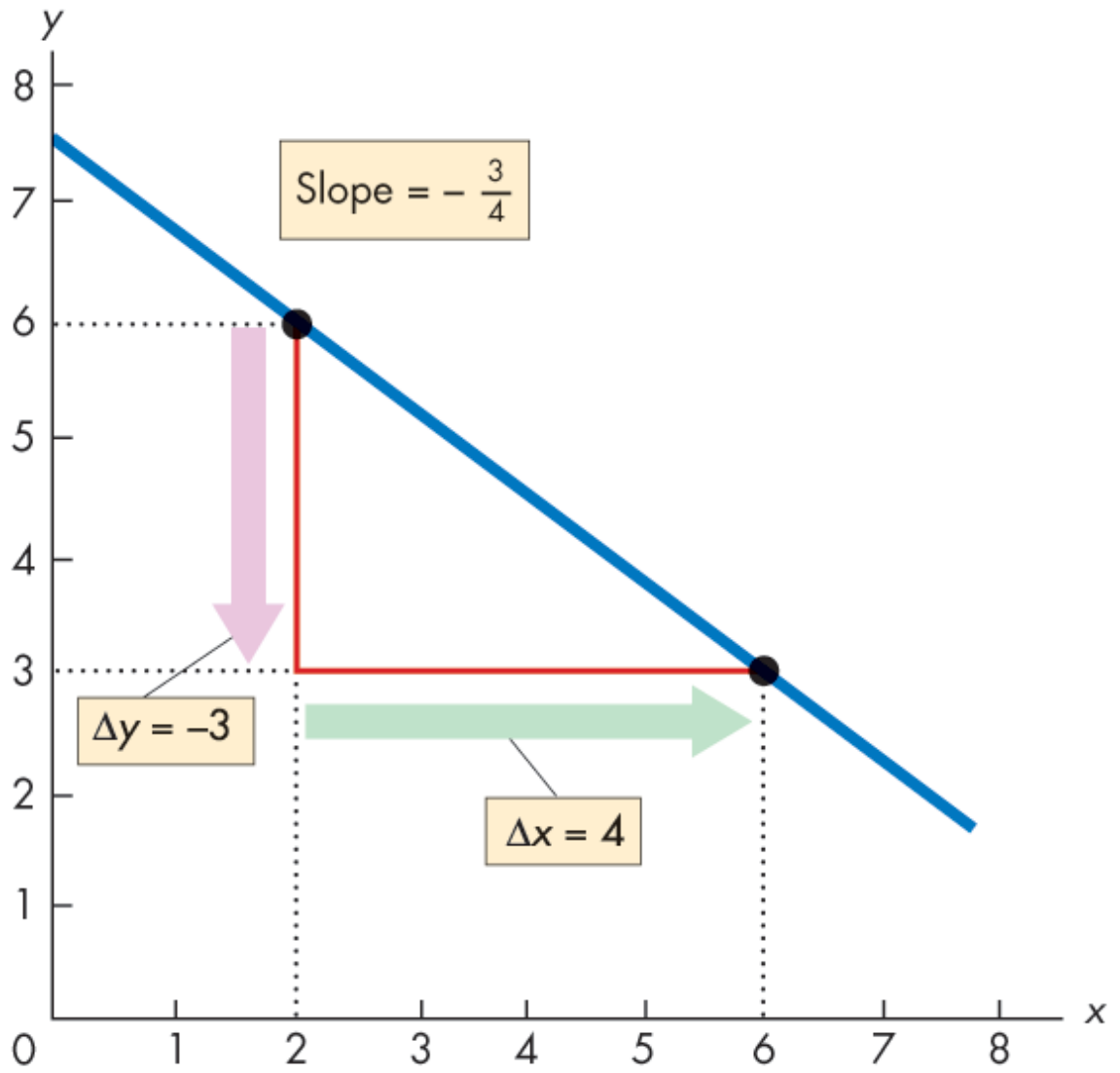
The Slope of a Relationship

The slope is negative if the line is downward sloping.



(b) Negative slope

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(b) Negative slope

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The Slope of a Relationship

The Slope of a Curved Line

The slope of a curved line at a point varies depending on where along the curve it is calculated.

We can calculate the slope of a curved line either at a point or across an arc.

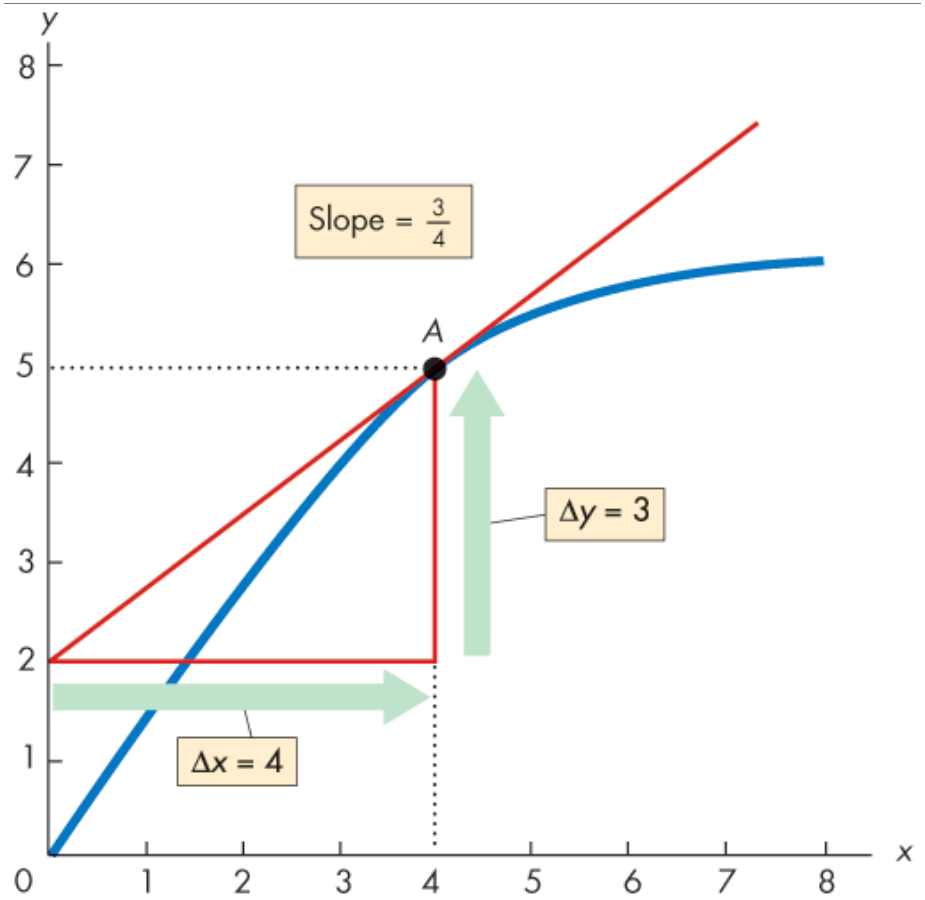
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The Slope of a Relationship

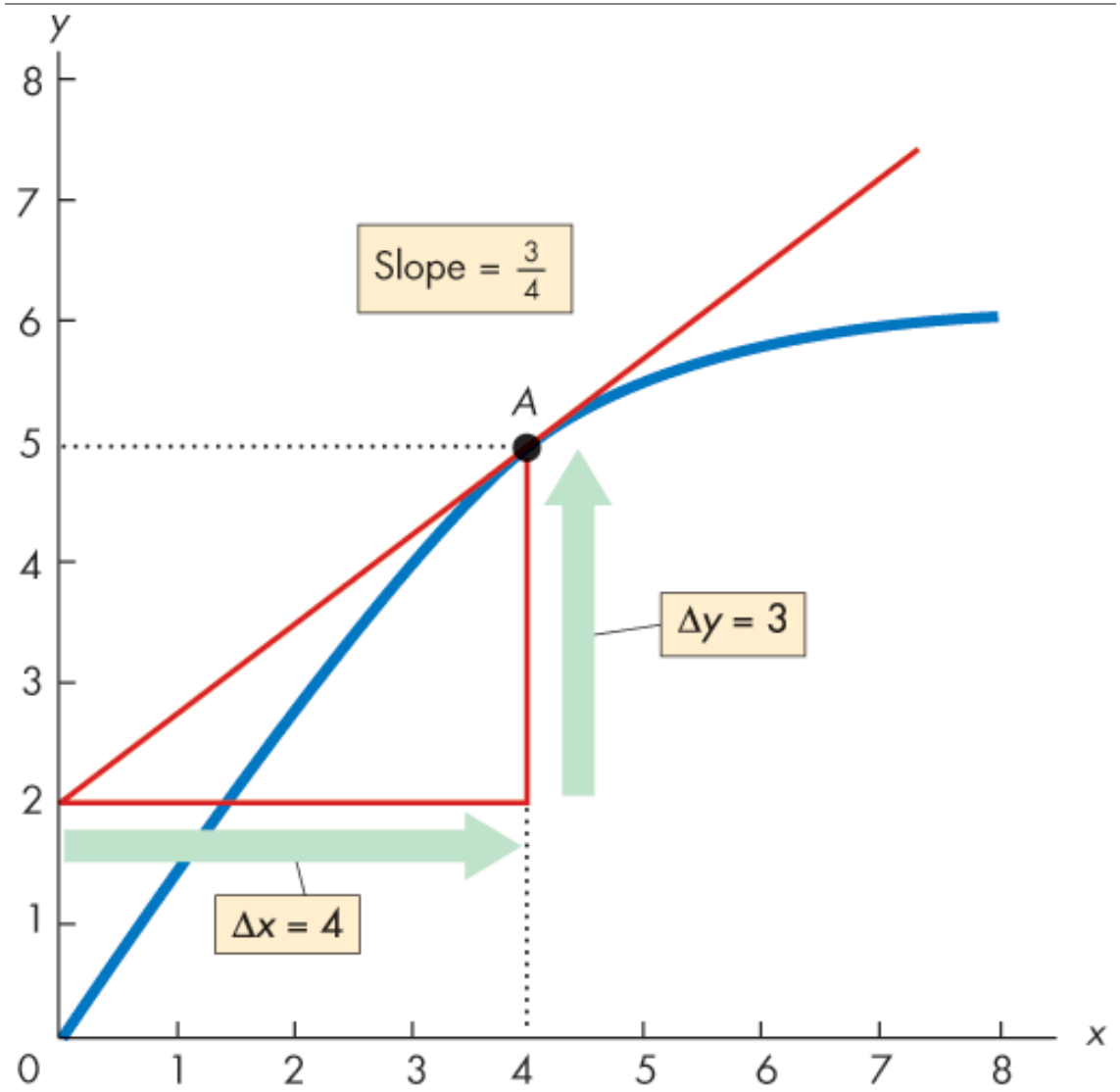
Slope at a Point

The slope of a curved line at a point is equal to the slope of a straight line that is the tangent to that point.

Here, we calculate the slope of the curve at point *A*.



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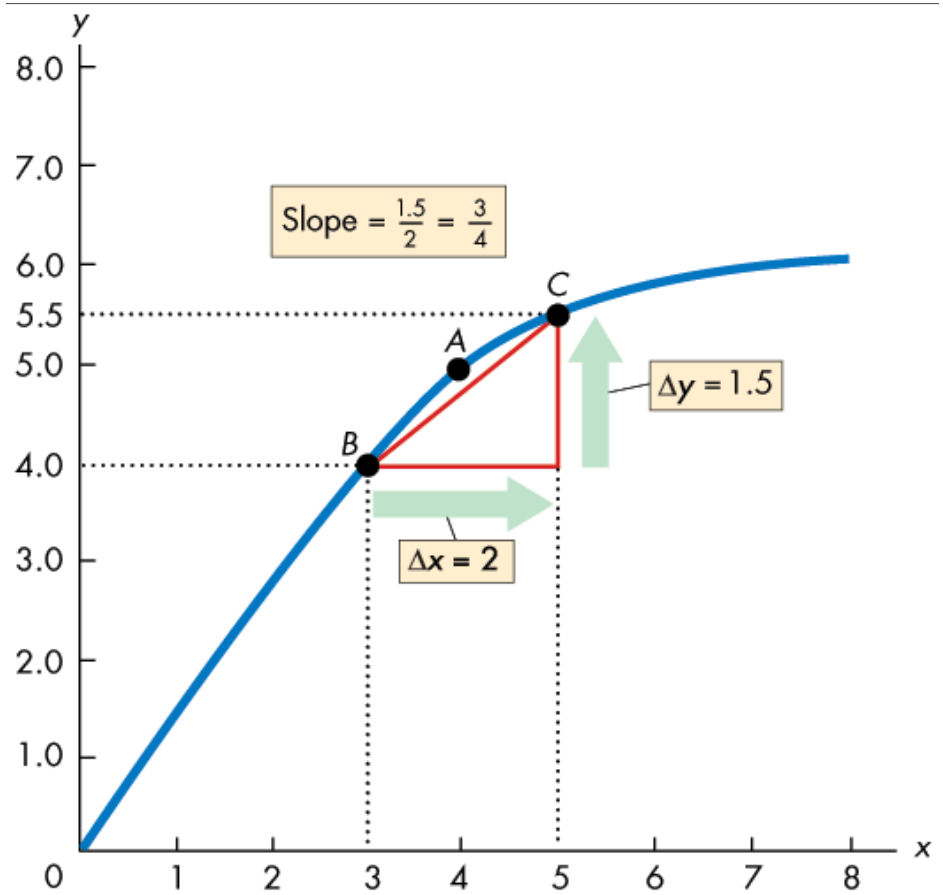
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The Slope of a Relationship

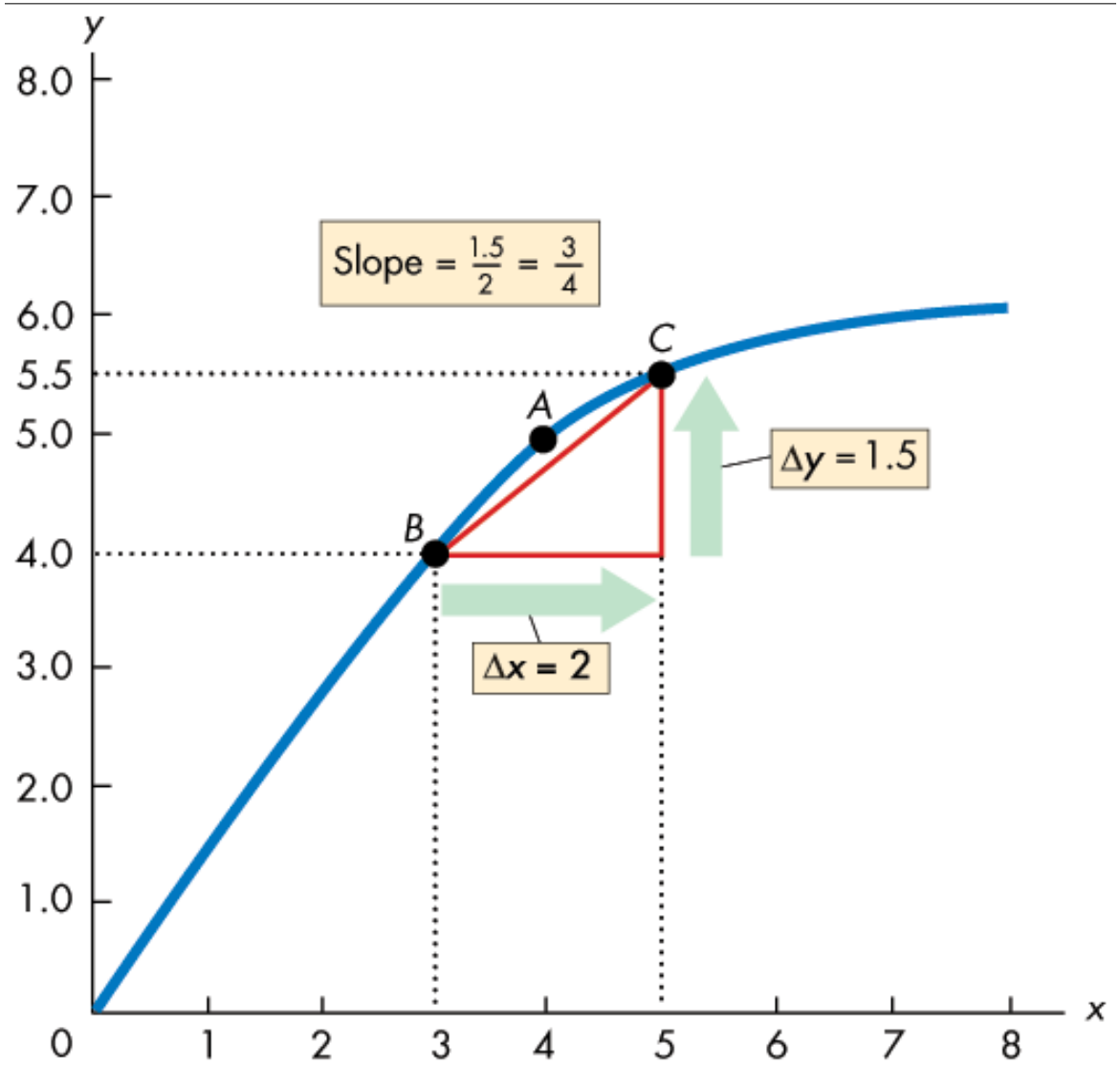
Slope Across an Arc

The *average* slope of a curved line across an arc is equal to the slope of a straight line that joins the endpoints of the arc.

Here, we calculate the average slope of the curve along the arc *BC*.



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Graphing Relationships Among More Than Two Variables

When a relationship involves more than two variables, we can plot the relationship between two of the variables by holding other variables constant—by using *ceteris paribus*.

Ceteris paribus

Ceteris paribus means "if all other relevant things remain the same."

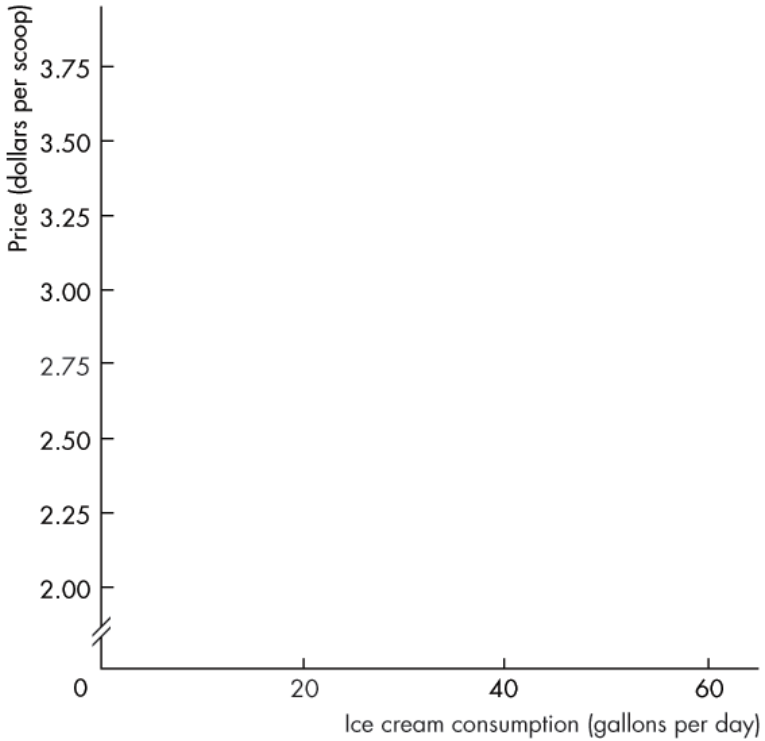
Figure A1.12 shows a relationship among three variables.

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Graphing Relationships Among More Than Two Variables

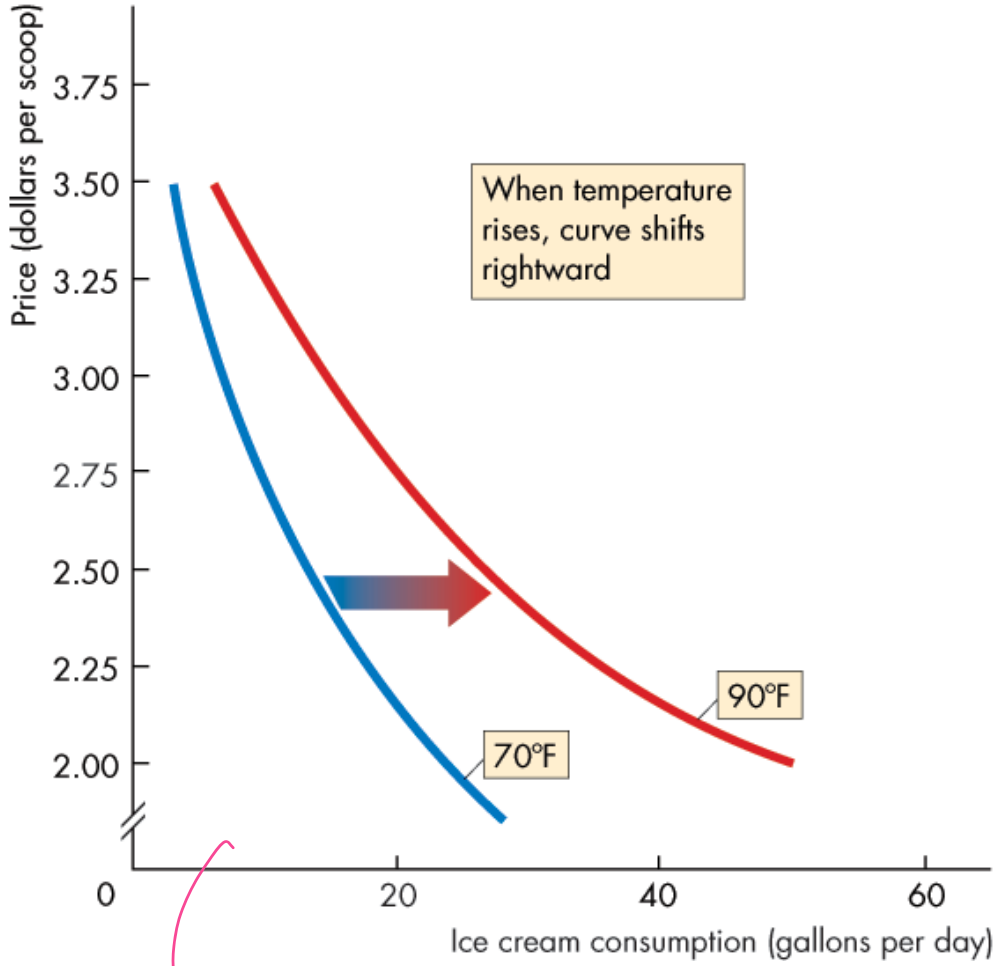
The table gives the quantity of ice cream consumed at different prices as the temperature varies.

Price (dollars per scoop)	Ice cream consumption (gallons per day)	
	70°F	90°F
2.00	25	50
2.25	18	36
2.50	13	26
2.75	10	20
3.00	7	14
3.25	5	10
3.50	3	6



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Price (dollars per scoop)	Ice cream consumption (gallons per day)	
	70°F	90°F
2.00	25	50
2.25	18	36
2.50	13	26
2.75	10	20
3.00	7	14
3.25	5	10
3.50	3	6



covered in the demand chapter

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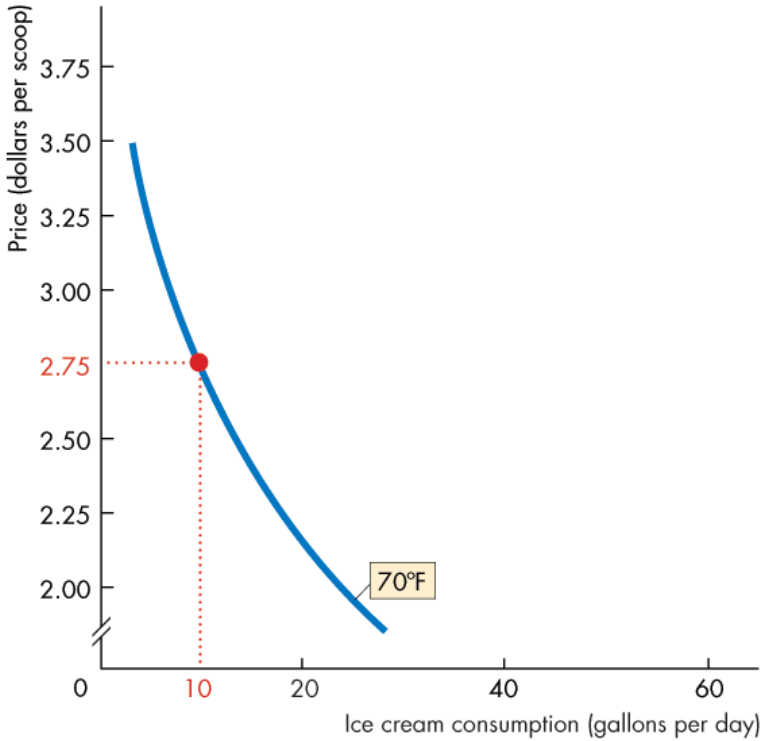
Graphing Relationships Among More Than Two Variables

To plot this relationship, we hold the temperature at 70°F.

At $2.75 a scoop, 10 gallons are consumed.

When the temperature is 70°F and the price changes, there is a movement along the blue curve.

Price (dollars per scoop)	Ice cream consumption (gallons per day)	
	70°F	90°F
2.00	25	
2.25	18	
2.50	13	
2.75	**10**	
3.00	7	
3.25	5	
3.50	3	



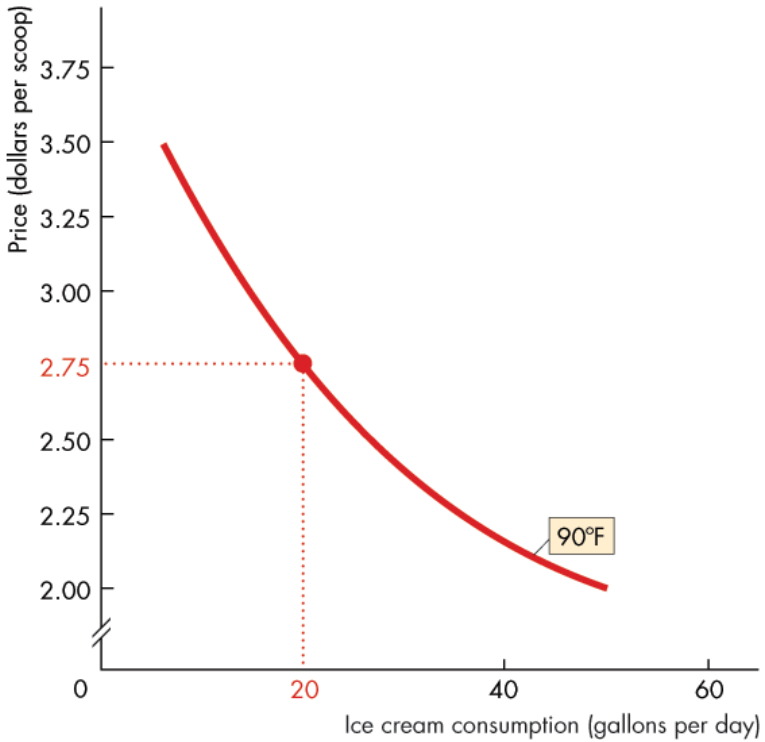
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Graphing Relationships Among More Than Two Variables

We can also plot this relationship by holding the temperature constant at 90°F.

At $2.75 a scoop, 20 gallons are consumed.

Price (dollars per scoop)	Ice cream consumption (gallons per day) 70°F	90°F
2.00		50
2.25		36
2.50		26
2.75		**20**
3.00		14
3.25		10
3.50		6

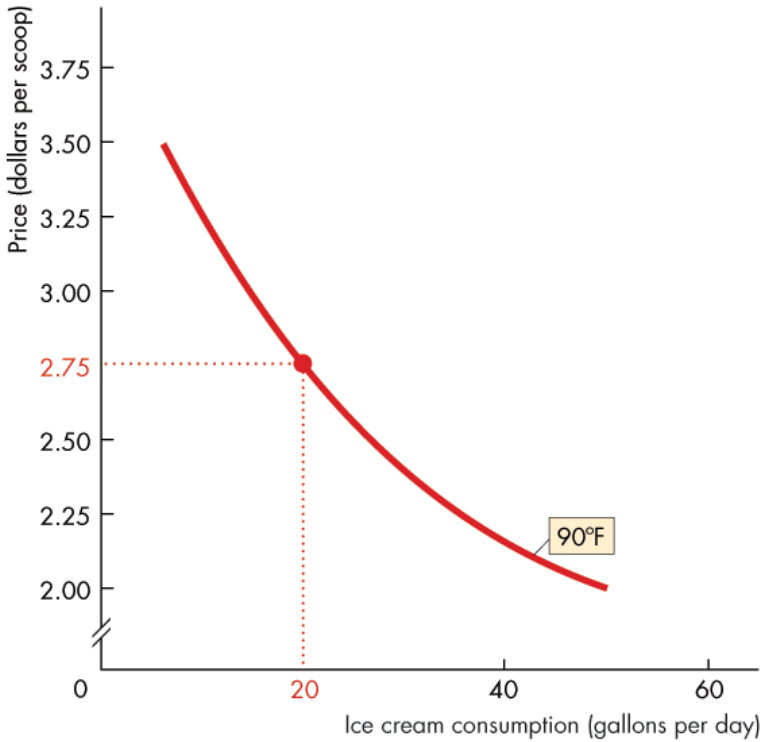


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Graphing Relationships Among More Than Two Variables

When the temperature is 90°F and the price changes, there is a movement along the red curve.

Price (dollars per scoop)	Ice cream consumption (gallons per day)	
	70°F	90°F
2.00		50
2.25		36
2.50		26
2.75		**20**
3.00		14
3.25		10
3.50		6



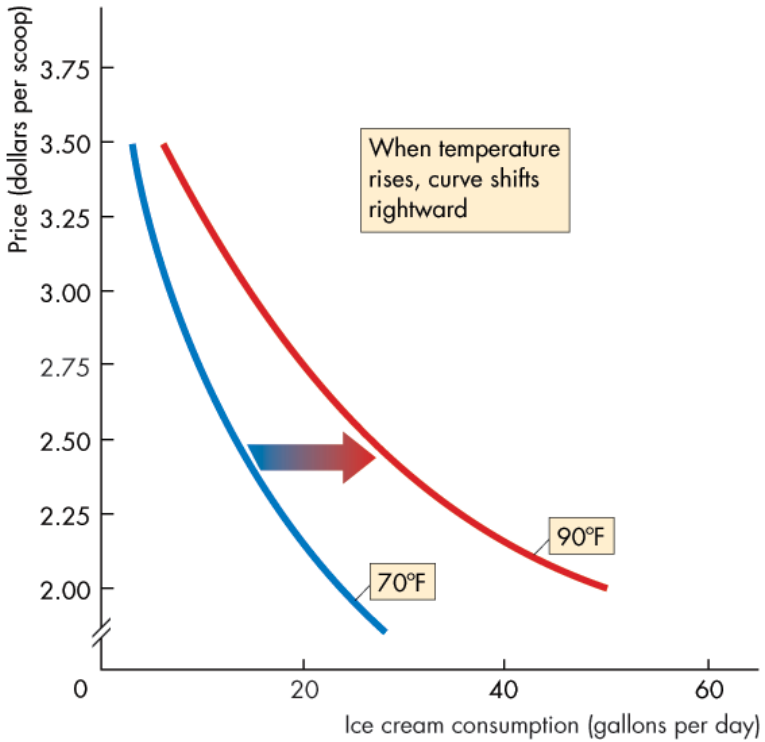
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Graphing Relationships Among More Than Two Variables

When Other Things Change

When the temperature rises from 70°F to 90°F, the curve showing the relationship shifts rightward from the blue curve to the red curve.

Price (dollars per scoop)	Ice cream consumption (gallons per day)	
	70°F	90°F
2.00	25	50
2.25	18	36
2.50	13	26
2.75	10	20
3.00	7	14
3.25	5	10
3.50	3	6



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